

CARD8 inflammasome activation during HIV-1 cell-to-cell transmission

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eLife Assessment

In this study, the authors investigated whether HIV-1 cell-to-cell transmission activates the CARD8 inflammasome in macrophages. The data **convincingly** support the idea that CARD8 is activated by the viral protease, promoting inflammation. The study's significance is further enhanced by including time-course analyses in primary T cells and macrophages and provides **valuable** insights into the role of CARD8 in HIV-induced inflammation.

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Abstract

Our previous work demonstrated that CARD8 detects HIV-1 infection by sensing the enzymatic activity of the HIV protease, resulting in CARD8-dependent inflammasome activation (Kulsuptrakul et al., 2023). CARD8 harbors a motif in its N-terminus that functions as a HIV protease substrate mimic, permitting innate immune recognition of HIV-1 protease activity, which when cleaved by HIV protease triggers CARD8 inflammasome activation. Here, we sought to understand CARD8 responses in the context of HIV-1 cell-to-cell transmission via a viral synapse. We observed that cell-to-cell transmission of HIV-1 between infected T cells and primary human monocyte-derived macrophages induces CARD8 inflammasome activation in a manner that is dependent on viral protease activity and largely independent of the NLRP3 inflammasome. Additionally, to further evaluate the viral determinants of CARD8 sensing, we tested a panel of HIV protease inhibitor resistant clones to establish how variation in HIV protease affects CARD8 activation. We identified mutant HIV-1 proteases that differentially cleave and activate CARD8 compared to wildtype HIV-1, thus indicating that natural variation in HIV protease affects not only the cleavage of the viral Gag-Pol polyprotein but also likely impacts innate sensing and inflammation.

Introduction

HIV-1 disease progression is characterized by chronic inflammation, immune activation, CD4⁺ T cell depletion and eventual destruction of the immune system and susceptibility to opportunistic infections. The primary cellular targets of HIV-1 are activated CD4⁺ T helper cells, specialized CD4⁺ T cell subtypes such as Th17 cells (Brenchley et al., 2008 [↗](#); Gosselin et al., 2009 [↗](#); Rodriguez-Garcia et al., 2014 [↗](#)), central memory cells (Chun et al., 1997b [↗](#), 1997a [↗](#), 1995 [↗](#)), and macrophages (Collman et al., 1990 [↗](#), 1989 [↗](#)). Chronic immune activation is primarily caused by rapid depletion of mucosal Th17 cells responsible for maintaining gut epithelial barrier integrity (Brenchley et al., 2008 [↗](#), 2006 [↗](#)). In addition to inflammation induced by circulating microbial ligands, inflammation can also originate from HIV-infected cells through activation of innate immune sensors that form cytosolic immune complexes known as inflammasomes. Inflammasome activation ultimately results in activation of pro-inflammatory caspases including caspase 1 (CASP1). Active CASP1 processes inflammatory cytokines and activates the pore-forming protein gasdermin D (GSDMD), which forms small pores in the plasma membrane and initiates a lytic form of cell death known as pyroptosis and the release of mature inflammatory cytokines interleukin (IL)-1 β and IL-18 (Broz and Dixit, 2016 [↗](#); Fink and Cookson, 2005 [↗](#)).

In prior work, we and others showed that the inflammasome-forming sensor CARD8 senses HIV-1 infection through the detection of HIV-1 protease (HIV^{PR}) activity (Clark et al., 2022 [↗](#); Kulsuptrakul et al., 2023 [↗](#); Wang et al., 2021 [↗](#)). While the canonical function of HIV^{PR} is to cleave viral polyproteins during virion maturation, active HIV^{PR} is also released into the host cell, which is sensed by CARD8 via HIV^{PR} cleavage of its N-terminus and subsequent inflammasome activation. In this way, the CARD8 N-terminus functions as a “molecular tripwire” to recognize the enzymatic activity of HIV^{PR} and other viral proteases (Castro and Daugherty, 2023 [↗](#); Nadkarni et al., 2022 [↗](#); Tsu et al., 2023 [↗](#)). Moreover, HIV^{PR} cleavage of CARD8 occurs rapidly after infection such that HIV^{PR} inhibitors and fusion inhibitors, but not reverse transcriptase (RT) inhibitors can prevent CARD8 inflammasome activation, implying that CARD8 detects HIV-1 viral protease activity of virion-packaged or “incoming” HIV^{PR} upon virion fusion (Kulsuptrakul et al., 2023 [↗](#); Wang et al., 2024 [↗](#), 2021 [↗](#)). Interestingly, CARD8 inflammasome activation in resting CD4⁺ T cells results in pyroptosis but not the release of pro-inflammatory cytokines IL-1 β or IL-18 (Wang et al., 2024 [↗](#)), suggesting that CARD8 inflammasome activation in T cells does not directly contribute to chronic inflammation. Here, we address whether or not CARD8 may influence HIV-1 pathogenesis through the maturation and release of IL-1 β from infected macrophages.

HIV-1 can be transmitted from one cell to another via two main mechanisms: “cell-free” infection through binding of free HIV-1 virions to target cells, and cell-to-cell infection whereby infected cells directly transfer virus to an uninfected target cell via the formation of a transient viral synapse (Chen et al., 2007 [↗](#); Galloway et al., 2015 [↗](#); Iwami et al., 2015 [↗](#)). Cell-to-cell transmission of HIV-1 has been reported between multiple HIV-1 target cell types including between active and resting CD4⁺ T cells (Agosto et al., 2018 [↗](#); Martin et al., 2010 [↗](#)) and between CD4⁺ T cells and macrophages (Baxter et al., 2014 [↗](#); Dupont and Sattentau, 2020 [↗](#); Lopez et al., 2019 [↗](#)). Cell-to-cell transmission delivers a large influx of virus to target cells, resulting in a high multiplicity of infection (MOI) (Agosto et al., 2015 [↗](#); Del Portillo et al., 2011 [↗](#); Duncan et al., 2013 [↗](#); Russell et al., 2013 [↗](#)), which has been proposed to enhance viral fitness by overwhelming host restriction factors including Tetherin/BST-2 (Jolly et al., 2010 [↗](#); Zhong et al., 2013 [↗](#)), SAMHD1 (Xie et al., 2019 [↗](#)), and TRIM5 α (Richardson et al., 2008 [↗](#)), and evading adaptive immune responses including broadly neutralizing antibodies (Abela et al., 2012 [↗](#); Dufloo et al., 2018 [↗](#)). Cell-to-cell spread of HIV-1 is thus an important consideration in studying CARD8 inflammasome activation.

Here, we investigated both host and viral determinants of CARD8 inflammasome activation upon HIV-1 infection. We evaluated CARD8 sensing of HIV^{PR} during cell-to-cell transmission of HIV-1 from T cell lines to myeloid cells in both immortalized and primary cell models of infection. We found that CARD8 inflammasome activation occurs in the context of cell-to-cell transmission from both SUPT1 cells, a T cell lymphoma cell line, to THP-1 cells, an acute myeloid leukemia cell line, and from primary CD4⁺ T cells to primary monocyte-derived macrophages. We also observed that HIV-triggered CARD8 inflammasome activation is largely independent of the NLRP3 inflammasome, which has previously been implicated in innate sensing of HIV-1 (Bandera et al., 2018 [↗](#); Chivero et al., 2017 [↗](#); Hernandez et al., 2013 [↗](#); Leal et al., 2020 [↗](#); Mamik et al., 2017 [↗](#); Zhang et al., 2021 [↗](#)). Our findings suggest that CARD8 sensing of HIV^{PR} activity during cell-to-cell transmission of HIV-1 to macrophages, leading to robust secretion of IL-1 β , may be a source of inflammatory cytokines that promote pathogenic chronic inflammation and disease progression. In addition, we also show that natural variation in HIV^{PR} due to resistance to protease inhibitors also affects CARD8 cleavage and subsequent inflammasome activation. Our results extend the role of incoming HIV^{PR} on CARD8-dependent inflammasome activation of inflammasome responses as a function of cell type, mode of transmission, and virus evolution in response to antiviral therapy.

Results

Cell-to-cell transmission of HIV-1 induces CARD8 inflammasome activation

Our previous work investigating HIV-dependent CARD8 inflammasome activation used the cationic polymer DEAE-dextran, which is a common reagent used to enhance viral infection in cell culture (Bailey et al., 1984 [↗](#)). However, we found that DEAE-dextran could induce inflammasome activation in the absence of viral infection in some “wildtype” (WT) THP-1 cell stocks (see **Supplemental Note**). These results prompted us to establish other models of HIV-1 infection and subsequent inflammasome activation that lack cationic polymers. Thus, we designed an *in vitro* coculture infection system to mimic HIV-1 cell-to-cell transmission by infecting SUPT1 cells, a T-cell lymphoma line (i.e., donor cells) and then mixing them with uninfected THP-1 cells (i.e., target cells). We opted for SUPT1 cells as the viral producer cell line because they are permissive to HIV-1 infection, and unlike THP-1 cells, SUPT1 cells do not respond to a known CARD8 inflammasome activator, ValboroPro (VbP), as assayed by both IL-1 β secretion and cell death, indicating that SUPT1 cells do not have a functional CARD8 inflammasome pathway (**Figure 1A** [↗](#)). This allowed us to infer that inflammasome outputs (e.g. IL-1 β secretion) in our coculture system occur upon cell-to-cell transmission of HIV-1 from SUPT1 cells to the CARD8-competent THP-1 cells.

We found that coculture of THP-1 cells with HIV-1_{LAI}-infected SUPT1 cells (**Figure 1B** [↗](#)) but not mock-infected SUPT1 cells results in robust inflammasome activation as indicated by IL-1 β secretion, suggesting that our coculture system, which lacks DEAE-dextran, can induce HIV-dependent inflammasome activation via cell-to-cell infection (**Figure 1C** [↗](#)). To further test this assumption, we prevented cell-to-cell contact using a virus-permeable transwell with a 0.4 μ m pore insert (**Figure 1B** [↗](#), left). We verified that there were equivalent amounts of infectious virus in the cell-to-cell condition versus the lower chamber of the transwell condition by measuring RT activity in the supernatant (**Figure 1B** [↗](#), right). Despite equivalent amounts of infectious virus in both conditions, we observed that HIV-1_{LAI}-infected SUPT1 cells (upper chamber) cocultured with THP-1 cells (lower chamber) did not lead to detectable IL-1 β secretion (**Figure 1C** [↗](#)). These data suggest that our SUPT1:THP-1 coculture system can trigger HIV-dependent inflammasome activation in a manner dependent on cell-to-cell contact.

We next assessed the role of CARD8 and other inflammasome sensors during cell-to-cell transmission of HIV-1. We cocultured mock- or HIV-1_{LAI}-infected SUPT1 cells with either WT or CARD8 KO THP-1 cells and compared inflammasome activation by measuring levels of secreted IL-

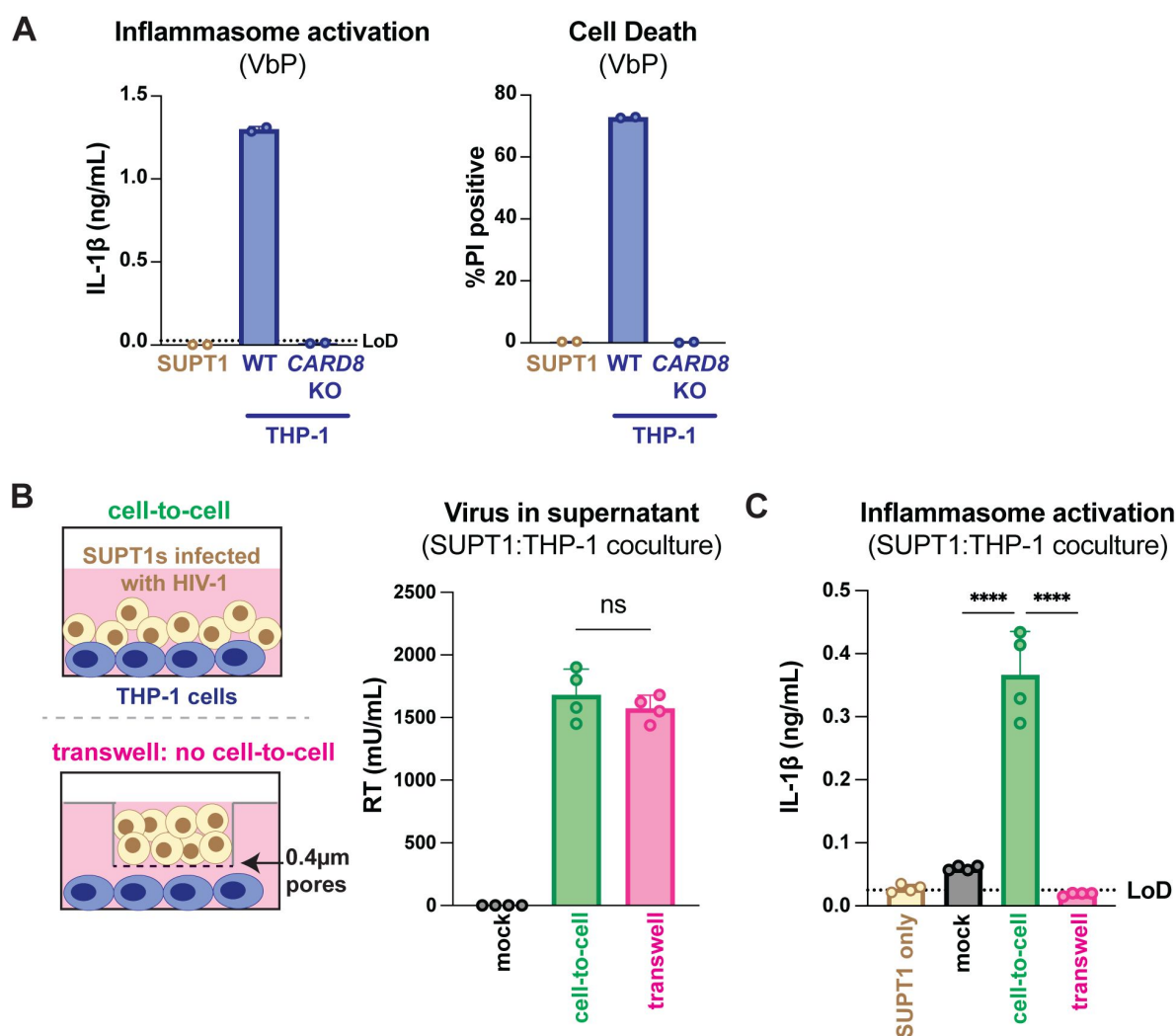


Figure 1.

HIV-1 cell-to-cell infection induces inflammasome activation.

(A) SUPT1 or THP-1 cells were primed with Pam3CSK4 (500ng/mL) overnight then treated with 5 μ M ValboraPro (VbP) for 24 hours then assessed for IL-1 β secretion and cell death via propidium iodide (PI) uptake. %PI positive was normalized to mock control. (B) (left) Schematic illustrating experimental setup for SUPT1:THP-1 cell coculture either with (bottom) or without (top) a transwell. (right) SUPT1 cells were either mock infected or infected with HIV-1_{LAI} then cocultured with primed WT THP-1 cells 20 hours post infection. Mock- or HIV-1_{LAI}-infected SUPT1 cells were either mixed with the THP-1 cells or put in a transwell with a virus-permeable membrane as shown in panel B (left). Supernatant in the cell-to-cell condition and in the supernatant outside of the transwell were sampled and measured for infectious HIV virions via reverse transcriptase (RT) assay or (C) IL-1 β secretion 3 days after starting the coculture. Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (A: n=2; B,C n=4 biological replicates). One-way ANOVA with (B) Tukey's or (C) Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

1 β . HIV-1_{LAI}-infected SUPT1 cells cocultured with WT but not *CARD8* KO THP-1 cells resulted in a significant increase in IL-1 β (**Figure 2A**). These results suggest that *CARD8* is the primary sensor that drives inflammasome activation in HIV-1 cell-to-cell transmission to THP-1 cells. Since the NLRP3 inflammasome has previously been implicated in HIV-dependent inflammasome activation (Bandera et al., 2018; Chivero et al., 2017; Hernandez et al., 2013; Leal et al., 2020; Mamik et al., 2017; Zhang et al., 2021), we also assessed the effects of the NLRP3 inflammasome-specific inhibitor MCC950 (Coll et al., 2015; Primiano et al., 2016) on inflammasome activation in our coculture system. Treatment with MCC950 or the caspase 1 (CASP1) inhibitor VX765 (Wannamaker et al., 2007) were sufficient to abrogate inflammasome activation induced by the ionophore nigericin, a well-characterized NLRP3 agonist (**Figure 2-figure supplement 1A**). However, in the HIV-1 coculture system, MCC950 treatment had only a modest effect on inflammasome activation while VX765 and the HIV^{PR} inhibitor lopinavir (LPV), which prevents *CARD8* cleavage by HIV-1^{PR} (Kulsuptrakul et al., 2023; Wang et al., 2021), completely abrogated IL-1 β secretion (**Figure 2B**). We observed similar results during HIV-1_{LAI} and HIV-1_{LAI}-VSVG cell-free infection of THP-1 cells in the presence of DEAE-dextran (**Figure 2-figure supplement 1B**). Taken together, these findings indicate that HIV-dependent inflammasome activation via cell-to-cell transmission is *CARD8*-dependent and largely NLRP3-independent.

CARD8, but not NLRP3, is required for inflammasome activation during HIV-1 cell-to-cell transmission into primary monocyte-derived macrophages

We next examined inflammasome activation upon HIV-1 cell-to-cell transmission in primary human monocyte-derived macrophages (MDMs). Previously, we had observed that *CARD8* could sense active HIV-1^{PR} released into the host cytosol following viral fusion, which we refer to as “incoming” HIV-1^{PR} in our cell-free infection system in THP-1 cells using DEAE-dextran and spinoculation (Kulsuptrakul et al., 2023). Thus, we assessed the importance of viral entry by coculturing MDMs from three independent donors with mock-, HIV-1_{LAI}-, or HIV-1_{NL4.3-BaL}-infected SUPT1 cells expressing *CCR5* (SUPT1-*CCR5*) and assayed for inflammasome activation (**Figure 3A**). HIV-1_{LAI} is a CXCR4 tropic strain unable to infect macrophages whereas HIV-1_{NL4.3-BaL} uses *CCR5* as a co-receptor which is a requirement for infection of macrophages. We observed inflammasome activation, as measured by IL-1 β secretion, in MDMs cocultured with HIV-1_{NL4.3-BaL}-infected SUPT1 cells but not in MDMs cocultured with mock- or HIV-1_{LAI}-infected SUPT1-*CCR5* cells (**Figure 3A**). This demonstrates that HIV-dependent inflammasome activation can occur in MDMs during cell-to-cell infection in a manner dependent on viral entry. To further ascertain if this inflammasome activation was *CARD8*-dependent and driven by incoming HIV^{PR} during SUPT1:MDM cell-to-cell transmission, we investigated the effects of different inhibitors on inflammasome activation in MDM cocultures with HIV-1_{NL4.3-BaL}-infected SUPT1s. We observed that IL-1 β secretion was abrogated by treatment with lopinavir, an HIV-1 protease inhibitor, and VX765, a CASP1 inhibitor, indicating that inflammasome activation in MDM cocultures is dependent on HIV^{PR} and CASP1, respectively (**Figure 3A**). In addition, we used an RT inhibitor, nevirapine (NVP), to prevent synthesis of *de novo* translated HIV^{PR}, and thus any *CARD8*-dependent IL-1 β secretion would only be due to incoming HIV^{PR} in the presence of NVP. Indeed, we observed HIV-dependent inflammasome activation in the presence of NVP that was added at the time of coculture, indicating that incoming HIV^{PR} is sufficient to elicit an inflammasome response (**Figure 3A**). Lastly, MDM cocultures treated with MCC950, an inhibitor of the NLRP3 inflammasome, had no effect on IL-1 β secretion (**Figure 3A**), and we observed similar inflammasome activation results regardless of whether the MDMs were primed with TLR1/2 agonist Pam3CSK4 versus TLR4 agonist lipopolysaccharide (LPS) (**Figure 3-figure supplement 1A**). Thus, cell-to-cell contact of infected cells with primary monocyte-derived macrophages can elicit an inflammasome response in a manner that is dependent on viral entry, CASP1, and incoming HIV^{PR}, and independent from NLRP3.

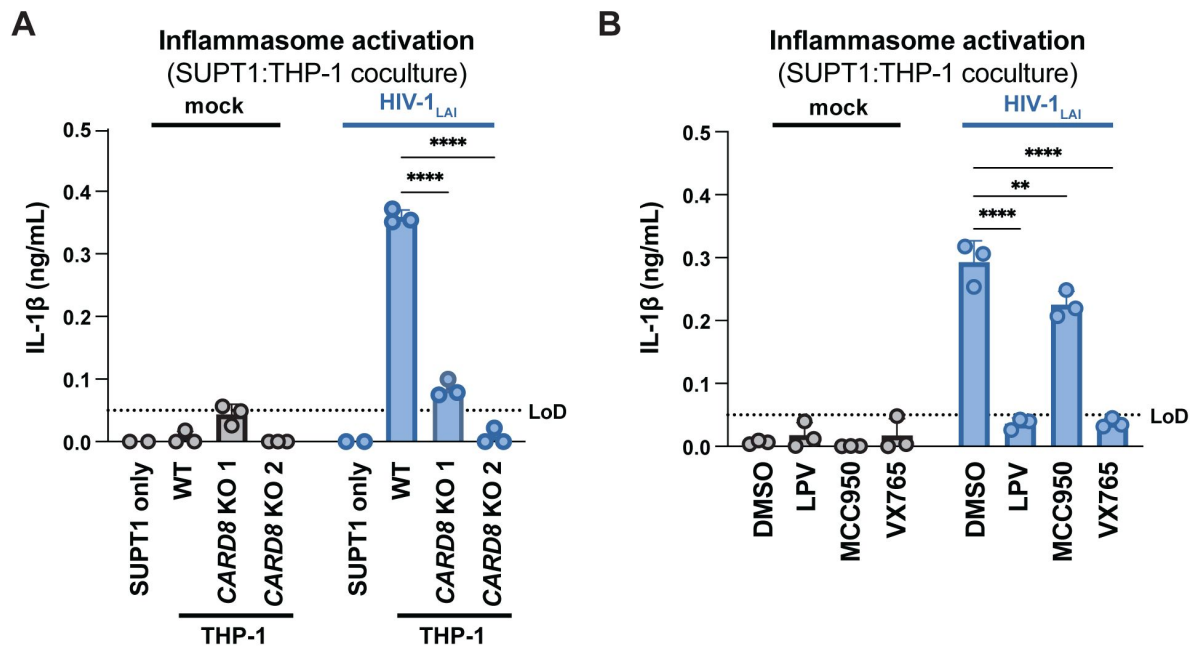


Figure 2.

HIV-1 cell-to-cell transmission induces CARD8-dependent activation largely independent of NLRP3.

(A) SUPT1 cells were either mock-infected or infected with HIV-1_{LAI} for 18-20 hours prior to coculture with wildtype (WT) or *CARD8* knockout (KO) THP-1 cells. The coculture was harvested 72 hours later to probe for IL-1 β secretion in the coculture supernatant via IL-1R reporter assay. THP-1 cells were primed with Pam3CSK4 (500ng/mL) for 16-24 hours prior to coculture. SUPT1 cells were infected with HIV-1_{LAI} such that 30% of the cells were positive for intracellular p24^{gag} after 18-20 hours. (B) SUPT1 cells were either mock or HIV-1_{LAI}-infected as in (A) for 18-20 hours then incubated in DMSO, lopinavir (LPV), MCC950, or VX765 at 0.01%, 5 μ M, 10 μ M, or 1 μ g/mL, respectively, for 15 minutes prior to coculturing with primed WT THP-1 cells. The coculture was assessed for subsequent inflammasome activation after 72 hours as in (A). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (n=3 biological replicates). Two-way ANOVA with Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

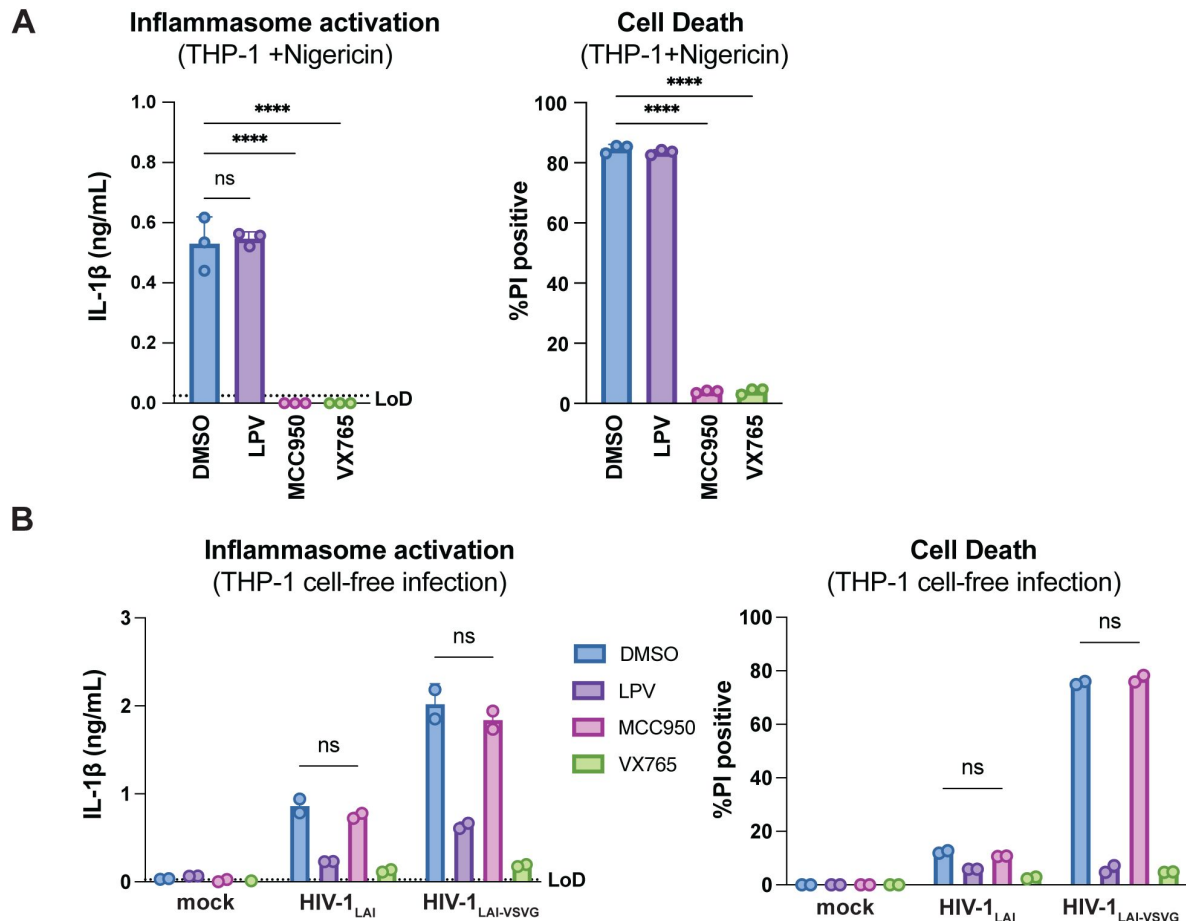


Figure 2- figure supplement 1.

HIV-dependent inflammasome activation is largely NLRP3-independent.

(A) Wildtype THP-1 cells were pre-treated with either DMSO, lopinavir (LPV), MCC950, or VX765 at 0.01%, 5 μ M, 10 μ M, or 1 μ g/mL, respectively, for 15 minutes prior to 4-hour treatment with 5 μ g/mL nigericin. Subsequent inflammasome activation was assessed via (A, left) IL-1 β secretion via IL-1R reporter assay and (A, right) cell death via propidium iodide (PI) dye uptake. (B) Wildtype THP-1 cells were pre-treated with indicated inhibitors as in (A) then infected with either HIV-1_{LAI} or VSV-G pseudotyped HIV-1_{LAI} (HIV-1_{LAI-VSVG}) in the presence of 10 μ g/mL DEAE-dextran such that both HIV-1_{LAI}-infected and HIV-1_{LAI-VSVG}-infected cells were ~30% positive for intracellular p24^{gag} after 24 hours by flow cytometry. Subsequent inflammasome activation was assessed 24 hours post infection via IL-1 β secretion and cell death as in (A). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (A: n=3, B: n=2 biological replicates). One-way (A) or Two-way (B) ANOVA with Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

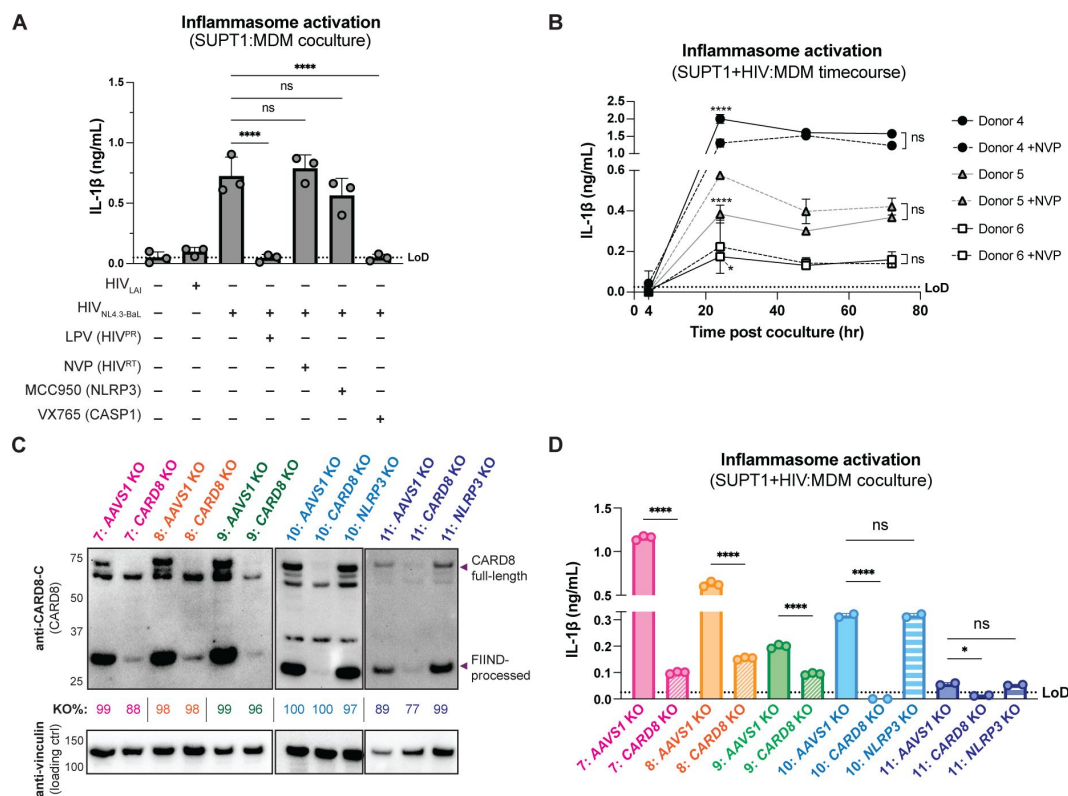


Figure 3.

Cell-to-cell HIV infection induces CARD8-dependent inflammasome activation in monocyte-derived macrophages (MDMs).

(A) MDMs from 3 independent donors were cocultured with SUPT1 cells expressing CCR5 (SUPT1-CCR5) that were mock-, HIV-1_{LAI}- or HIV-1_{NL4.3-BaL}-infected then assayed for inflammasome activation 48 hours post-coculture for IL-1 β secretion. Fifteen minutes before starting the coculture, SUPT1-CCR5 cells infected with HIV-1_{NL4.3-BaL} were pre-treated with either DMSO, lopinavir (5 μ M), nevirapine (50 μ M), MCC950 (10 μ M) or VX765 (1 μ g/mL), inhibiting HIV-1 protease (HIV^{PR}), HIV-1 reverse transcriptase (HIV^{RT}), NLRP3, or caspase 1 (CASP1), respectively. (B) MDMs from 3 independent donors were cocultured with SUPT1-CCR5 cells infected with HIV-1_{NL4.3-BaL} in either the presence or absence of nevirapine (NVP). Supernatant was harvested at 4, 24, 48, or 72 hours to assay for IL-1 β secretion. (C) MDMs from 5 independent donors were knocked out (KO) for AAVS1, CARD8, or NLRP3 using a Synthego gene KO kit then immunoblotted using an anti-CARD8 antibody or anti-vinculin. Full-length and FIIND-processed CARD8 intermediates are marked with a purple arrow. Table between CARD8 and vinculin blot shows Synthego gene KO% scores for each donor KO line. (D) AAVS1, CARD8 or NLRP3 KO MDM lines from (C) were primed with Pam3CSK4 (500ng/mL) overnight and then cocultured with SUPT1-CCR5 cells mock-, or HIV-1_{NL4.3-BaL}-infected then assayed for inflammasome activation 48 hours post-coculture for IL-1 β secretion. For all SUPT1:MDM experiments, SUPT1-CCR5 cells were infected with HIV-1_{LAI} or HIV-1_{NL4.3-BaL} such that 5-20% of cells were positive for intracellular p24^{ag} after 20 hours. IL-1 levels shown were normalized to the SUPT1 mock-infected coculture control. Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (A: n=3 independent donors, B: n=2 biological replicates for each donor, D: n=3 technical replicates per donor). One-way ANOVA with (A) Tukey's or (D) Sidak's test or (B) two-way ANOVA with Tukey's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

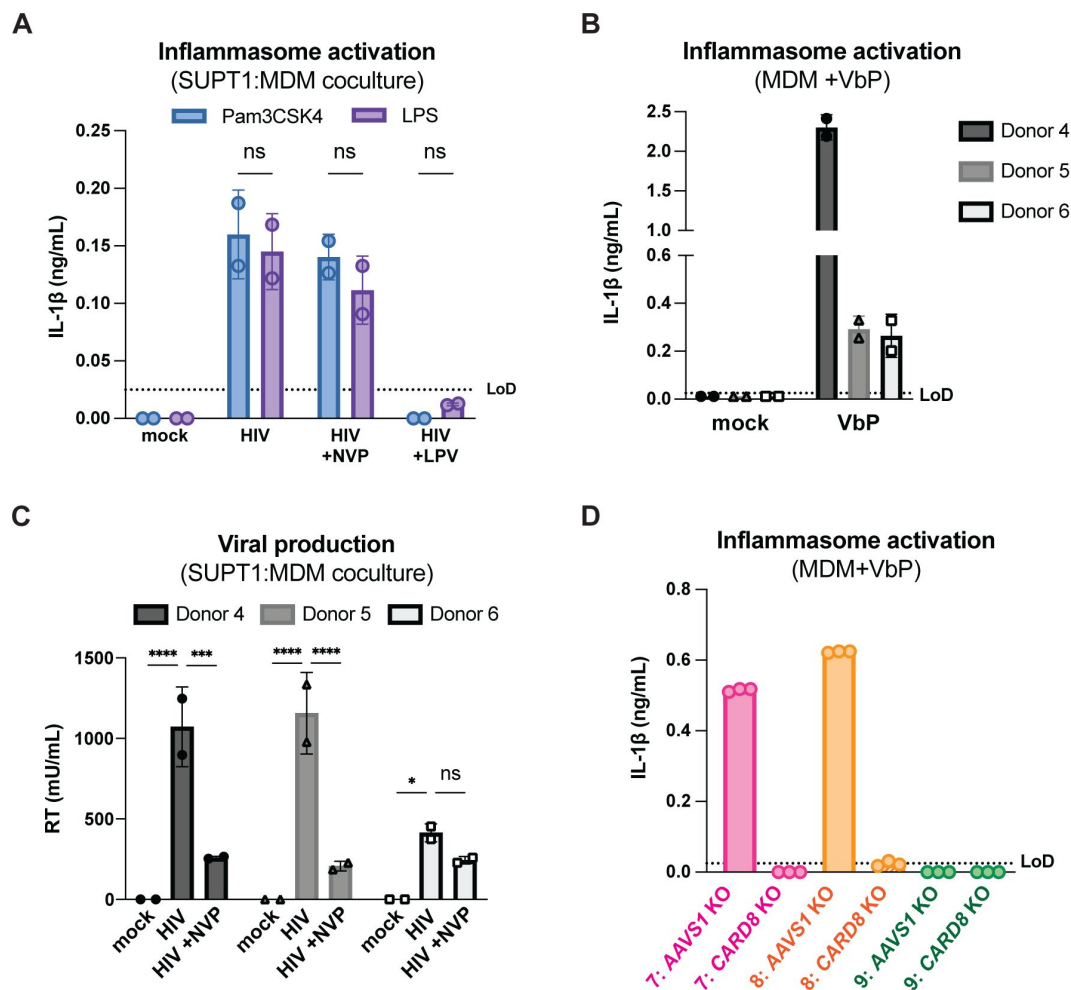


Figure 3-figure supplement 1

Cell-to-cell HIV infection induces CARD8-dependent activation in monocyte-derived macrophages (MDMs).

(A) MDMs from donor 6 were primed with either 500ng/mL Pam3CSK4 or 5μg/mL LPS then cocultured with SUPT1-CCR5 cells that had been infected with HIV-1_{NL4.3-BaL} 24 hours prior to coculture. Each coculture was started in the presence of DMSO, nevirapine (NVP) or lopinavir (LPV). Supernatant was harvested 72 hours post coculture to assay for IL-1β secretion. (B) MDMs from the same 3 independent blood donors assayed in [Figure 3B](#) were primed overnight with Pam3CSK4 then treated with 10μM VbP for 24 hours before assaying for IL-1β secretion via IL-1 reporter assay. (C) Supernatant from SUPT1:MDM coculture experiment done in [Figure 3B](#) was harvested at 48 hours post coculture to assay for infectious virions via reverse transcriptase (RT) assay. (D) AAVS1 or CARD8 KO MDMs from donor 7-9 assayed in [Figure 3C](#) were primed and treated with VbP for 24 hours then assayed for IL-1β secretion. IL-1 levels from VbP treatment were normalized to untreated mock control. Dotted line indicates limit of detection (LoD). Datasets represent mean ± SD (A: n=2 biological replicates for one donor, B-C: n=2 technical replicates for each independent donor, D: n=3 technical replicates per donor). Two-way ANOVA with (A) Sidak's or (C) Dunnett's test (using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001).

To further assess the timing of this inflammasome activation, we conducted a SUPT1:MDM time course coculture experiment with 3 additional donors, assaying IL-1 β secretion at 4, 24, 48, and 72 hours post coculture in the presence and absence of NVP. We observed inflammasome activation as determined by IL-1 β secretion by 24 hours post coculture in all donors that persisted at a similar level at 48 and 72 hours post coculture (**Figure 3B**). As expected, we observed donor-to-donor variation in the extent to which IL-1 β secretion occurred following HIV-1 infection. However, per donor, HIV-1-driven IL-1 β levels were comparable to that of VbP-induced inflammasome activation (**Figure 3-figure supplement 1B**). Moreover, adding NVP had no effect on IL-1 β secretion in HIV-1 infected MDM cocultures (**Figure 3B**). To verify that NVP was functional, we assayed the supernatant of the mock and NVP-treated cocultures from **Figure 3B** at 48 hours post-coculture for infectious virions via an assay for RT activity and observed lower RT activity in NVP-treated MDM donors (**Figure 3-figure supplement 1C**). Unlike the ‘cell-free’ infection conditions in which we previously observed an increase in IL-1 β levels 4h post-infection, we did not detect measurable differences in IL-1 β secretion at this early timepoint following the establishment of SUPT1:MDM coculture. Nevertheless, the data are consistent with incoming HIV^{PR} being responsible for inflammasome activation during cell-to-cell transmission of HIV because the induction of IL-1 β persists in the presence of NVP which would block any *de novo* synthesis of new *gag/pol* products (**Figure 3A** and **B**). Taken together, these data suggest that in the context of cell-to-cell transmission, CARD8 is likely the inflammasome-forming sensor that detects HIV-1 infection via incoming HIV^{PR} activity in primary monocyte-derived macrophages.

To specifically address the role of CARD8 in HIV-1 induced inflammasome activation in MDMs, we genetically edited MDMs by isolating monocytes from five donors and electroporating them with Cas9 RNPs complexed with three unique sgRNAs per gene targeting *AAVS1*, a safe harbor locus, *CARD8* or *NLRP3* (only 2 donors for *NLRP3* KO). Edited MDMs were then differentiated for 6 days prior to evaluating KO efficiency and initiating cocultures with HIV-1-infected SUPT1 cells. To verify KO efficiency, we immunoblotted with an antibody that detects the CARD8 C-terminus in *CARD8* KO MDMs relative to the *AAVS1* KO control and observed a marked reduction of the full-length and FIIND-processed CARD8 protein in all 5 donors (**Figure 3C**). In addition, we confirmed *AAVS1*, *CARD8*, and *NLRP3* KO at the genetic level via Synthego ICE analysis (Conant et al., 2022), measuring >85% KO efficiency (**Figure 3C**). We also observed robust inflammasome activation upon treatment with CARD8 inflammasome activator VbP as measured by IL-1 β secretion in *AAVS1* KO MDMs from 2 of the 3 donors, which was completely abrogated in *CARD8* KO MDMs, confirming functional loss of CARD8 (**Figure 3-figure supplement 1D**). We then cocultured either *AAVS1* KO, *CARD8* KO, or *NLRP3* KO MDMs with mock or HIV-1_{NL4.3-BaL}-infected SUPT1-CCR5 cells at a 1:1 ratio and measured inflammasome activation via IL-1 β secretion 48 hours post-coculture. In all 5 donors, we observed significant reduction in inflammasome activation in *CARD8* KO cocultures relative to the *AAVS1* KO control (**Figure 3D**). Consistent with our findings with SUPT1:THP-1 coculture, we observed no difference in inflammasome activation when coculturing infected SUPT1 cells with *AAVS1* KO vs *NLRP3* KO MDMs, suggesting that the inflammasome activation was also largely NLRP3-independent (**Figure 3D**). Taken together, these data demonstrate that CARD8 is required for inflammasome activation in MDMs during HIV-1 cell-to cell transmission.

Coculture of HIV-1 infected primary CD4⁺ T cells with primary MDMs elicits CARD8-dependent inflammasome activation

We next investigated inflammasome activation in the context of cell-to-cell infection using primary CD4⁺ T cells as donor cells, rather than SUPT1 cells, and primary MDMs as target cells. Mock or HIV-1_{NL4.3-BaL} infected CD4⁺ T cells were cocultured with MDMs in the presence or absence of various inhibitors (as in **Figure 3A**). We assessed inflammasome activation via IL-1 β secretion 72 hours post coculture and observed inflammasome activation when coculturing with HIV-infected T cells but not mock infected T cells or cocultures treated with LPV or VX765, demonstrating that inflammasome activation is driven by HIV^{PR} and CASP1 (**Figure 4A**) and

consistent with our findings using the SUPT1:MDM coculture (**Figure 3**). We also observed that inflammasome activation persisted in NVP-treated cocultures, suggesting that incoming protease is also important for inflammasome activation in the context of cell-to-cell transmission of HIV-1 from primary CD4⁺ T cells to primary MDMs (**Figure 4A**). To confirm the potency of LPV and NVP, we assayed for infectious virions in the supernatant of these CD4:MDM cocultures and detected a dramatic decrease in RT activity in the presence of either of these drugs, indicating that these drugs were efficacious at this dose (**Figure 4-figure supplement 1A**). We also conducted a time course experiment with a CD4:MDM coculture from an independent donor in the presence or absence of LPV. Similar to the SUPT1:MDM time course (**Figure 3B**), we were able to detect elevated levels of IL-1 β by 24 hours post coculture, which was again strictly dependent on the enzymatic activity of the viral protease as LPV treatment completely inhibited IL-1 β secretion (**Figure 4B**).

To interrogate the specific role of CARD8 in this primary CD4:MDM coculture system, we generated AAVS1 (as a control) or CARD8 KO MDMs and cocultured the MDMs with either mock or HIV-infected primary CD4⁺ T cells then assayed for inflammasome activation 48 hours post coculture. KO efficiency was confirmed via immunoblot and functional response to VbP (in one of the two donors) (**Figure 4C**, **Figure 4-figure supplement 1B**). We detected a significant decrease in inflammasome activation when infected T cells were cocultured with CARD8 KO versus AAVS1 KO MDMs (**Figure 4D**), indicating that CARD8 is required for inflammasome activation in MDMs during HIV-1 transmission from primary CD4⁺ T cells to MDMs. Taken together, our data indicate that CARD8 plays a pivotal role in sensing and responding to HIV-1 cell-to-cell infection between primary CD4 T cells and macrophages.

Protease inhibitor resistant strains of HIV-1 differentially cleave and activate CARD8

The consequences of CARD8 inflammasome activation on viral replication have been challenging to assess given that viral fitness is intrinsically linked to viral protease processing of the viral polyprotein such that inhibiting HIV^{PR} also prevents viral replication. In an attempt to circumvent this issue, we surveyed a panel of multi-HIV^{PR} inhibitor-resistant (PI-R) infectious molecular clones of HIV-1 (Varghese et al., 2013). This panel of PI-R molecular clones vary in resistance to HIV protease inhibitors including nelfinavir (NFV), fosamprenavir (FPV), saquinavir (SQV), indinavir (IDV), atazanavir (ATV), lopinavir (LPV), tipranavir (TPV), and darunavir (DRV). Each molecular clone encodes 4 to 11 mutations in HIV^{PR} as well as various compensatory HIV^{gag} mutations (Varghese et al., 2013) (**Table S1**).

We initially tested if PI-R HIV-1 proviruses differentially cleave CARD8 by co-transfecting HEK293T cells with an expression plasmid encoding an N-terminal mCherry tagged human CARD8 and either wildtype HIV-1_{LAI} or PI-R HIV-1 proviruses. HIV-1_{LAI} protease cleaves CARD8 between phenylalanine (F) 59 and F60 (Wang et al 2021), resulting in a ~33kDa product (**Figure 5A**, top). By quantifying the 33kDa CARD8 cleavage product with each HIV-1 provirus, we identified a PI-R clone that exhibited similar efficiency at cleaving CARD8 to HIV-1_{LAI} (i.e., PI-R1), PI-R clones that were markedly less efficient at cleaving CARD8 than HIV-1_{LAI} (i.e., PI-R2, PI-R3, PI-R5, PI-R9, and PI-R10) and two PI-R clones, PI-R12 and PI-R13, that were more efficient at cleaving CARD8 than HIV-1_{LAI} (**Figure 5A**, top, **Table S1**). Of note, all PI-R proviruses had similar levels of HIV^{PR} activity for HIV^{gag} polyprotein processing from p55^{gag} to p24^{gag} as indicated by the ratio of p24^{gag}/p55^{gag} quantified from the anti-p24^{gag} immunoblot (**Figure 5A**, middle). These results indicate that naturally occurring HIV-1 protease mutations can influence host targets like CARD8.

We next assessed if PI-R clones exhibiting reduced (PI-R2 and -9) or increased (PI-R12 and -13) cleavage of CARD8 relative to HIV-1_{LAI} (**Figure 5A** and **B**, and **Table S1**) resulted in differential inflammasome activation. HEK293T cells endogenously express CARD8 but lack the downstream components (i.e., CASP1, GSDMD, and IL-1 β /IL18) of the inflammasome pathway.

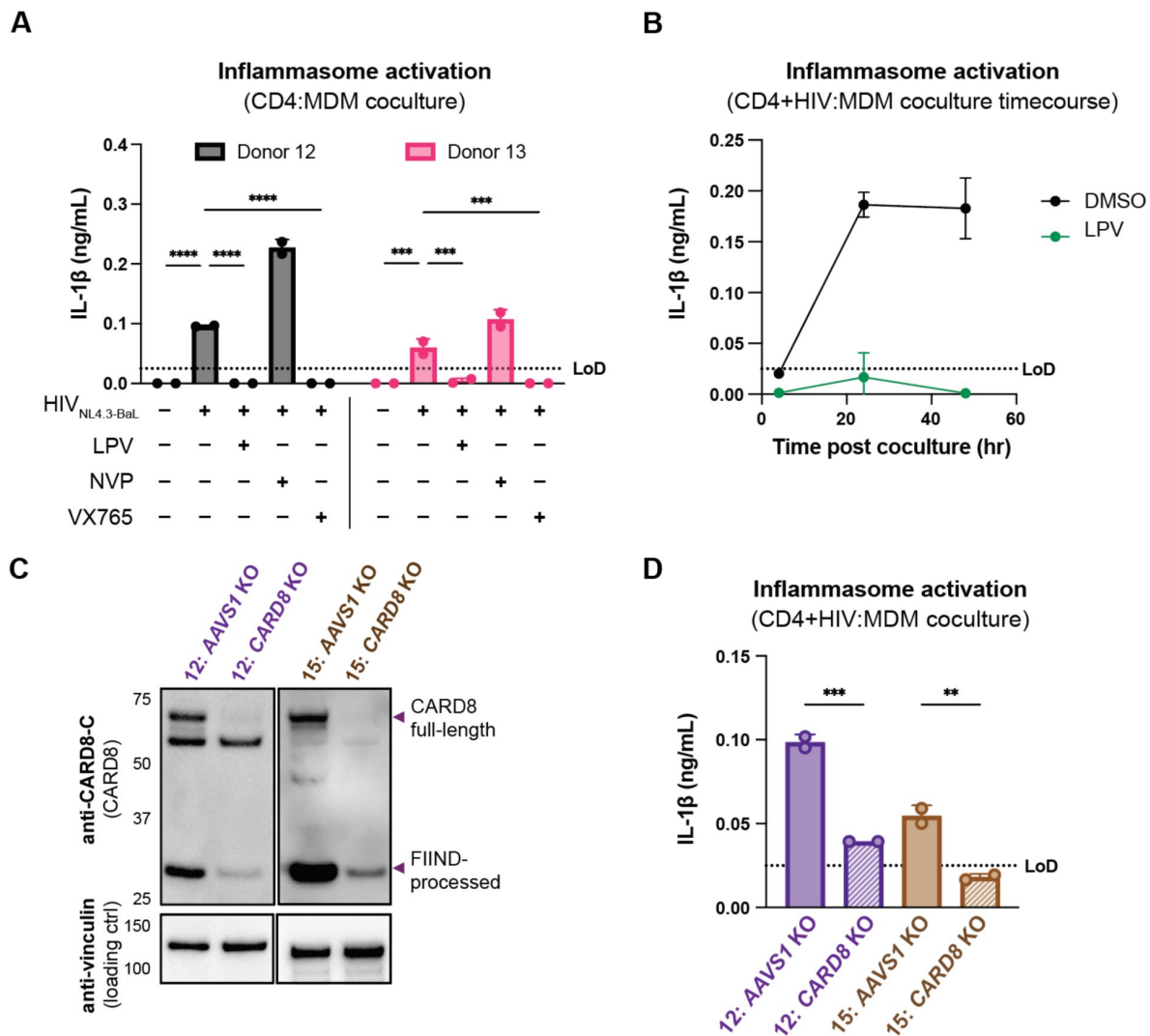


Figure 4.

Primary CD4+ T cell:MDM coculture elicits CARD8-dependent inflammasome activation.

(A) CD4+ T cells from a blood donor were isolated, activated, and either mock infected or infected with HIV-1_{NL4.3-BaL} for 3 days such that ~10% of cells were positive for intracellular p24^{gag}. MDMs were primed with Pam3CSK4 then cocultured with mock or HIV-1 infected primary CD4 T cells in the presence or absence of lopinavir (LPV), nevirapine (NVP), or VX765, inhibiting HIV protease, reverse transcriptase, or caspase 1, respectively. Supernatants were harvest 3 days post coculture to assay for IL-1 β secretion via IL-1 reporter assay. (B) CD4+ T cells from donor 12 and MDMs from donor 14 were cocultured as in A in the presence or absence of LPV. Supernatant was harvested at 4, 24 and 48 hours post coculture to probe for IL-1 β secretion. (C) AAVS1 or CARD8 MDM KOs were immunoblotted using an anti-CARD8 antibody or anti-vinculin. Full-length and FIIND-processed CARD8 intermediates are marked with a purple arrow. (D) AAVS1 or CARD8 KO MDMs from (C) were cocultured with CD4+ T cells infected with HIV-1_{NL4.3-BaL} then assayed for IL-1 β secretion 48 hours post coculture. The donor 12 cocultures consisted of autologous CD4s and MDMs whereas the MDMs from donors 13-15 were cocultured with donor 12 CD4s. Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (A,D: n=2 technical replicates for each donor, B: n=3 technical replicates for one donor). (A) Two-way ANOVA with Tukey's test (D) One-way ANOVA with Sidak's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

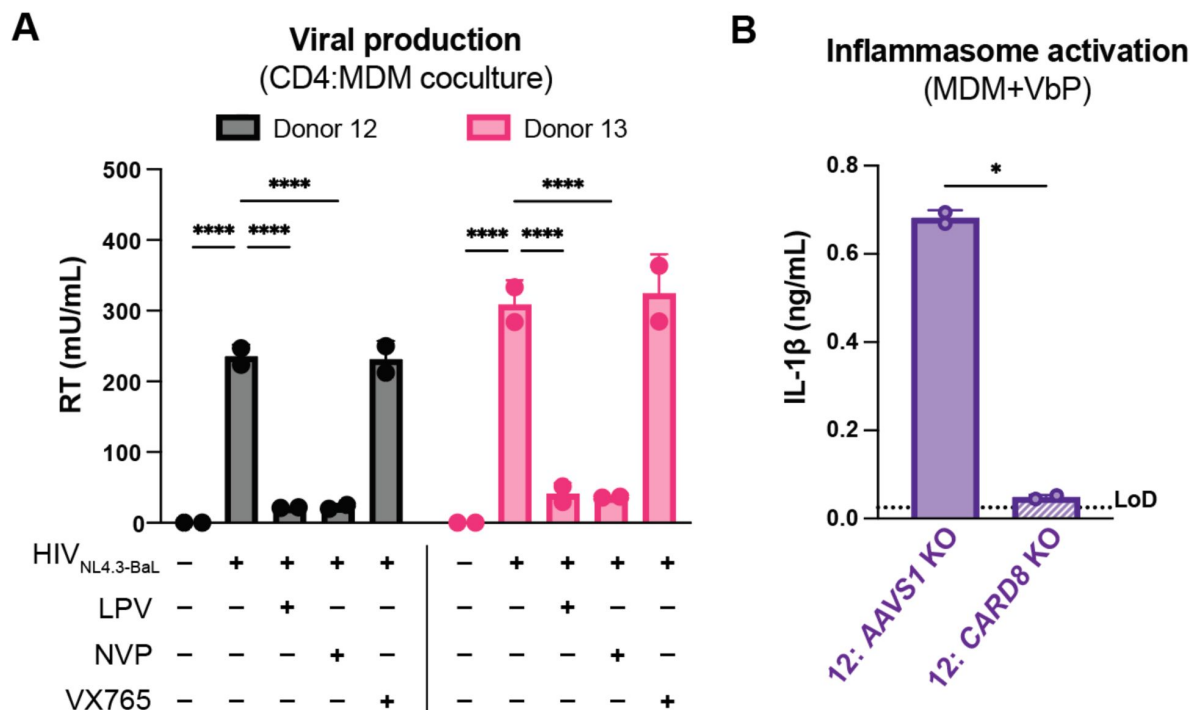


Figure 4-figure supplement 1.

Primary CD4 T cell:MDM coculture elicits CARD8-dependent inflammasome activation.

(A) Supernatant was harvested 72 hours post coculture from coculture described in [Figure 4A](#) then assayed for infectious virions via (RT) transcriptase assay. (B) AAVS1 or CARD8 KO MDMs from donor 12 were primed with Pam3CSK4 then treated with VbP for 24 hours and probed for IL-1β secretion. (n=2 technical replicates for each donor). (A) Two-way ANOVA with Tukey's test (B) One-way ANOVA with Sidak's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

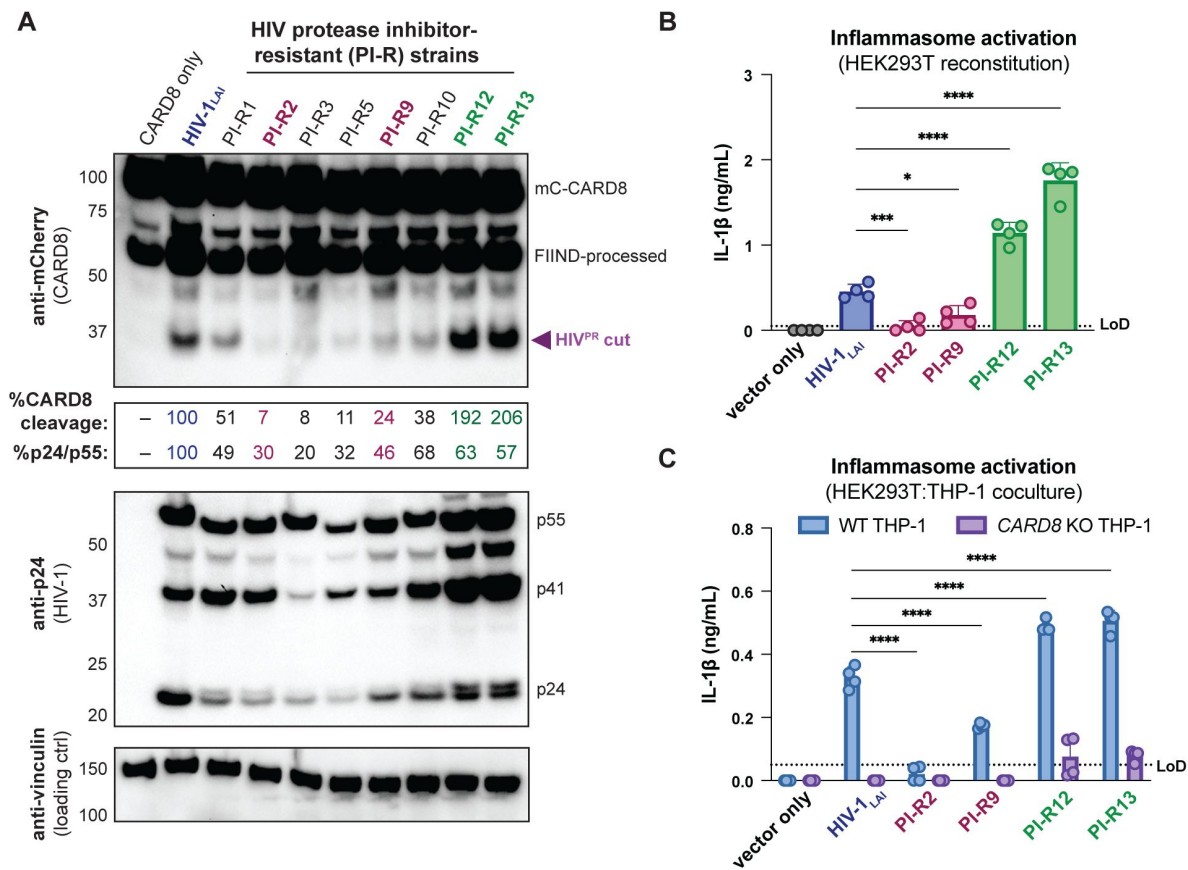


Figure 5.

Protease inhibitor resistant strains of HIV-1 differentially cleave and activate CARD8.

(A) HEK293T cells were transfected with a construct encoding CARD8 with an N-terminal mCherry tag (mCherry-CARD8) and indicated HIV-1 proviral constructs. Protease inhibitor-resistant (PI-R) clones of HIV-1 are a subset of a panel expressing prototypical multidrug resistant HIV-1 protease (HIV^{PR}) in an NL4.3 backbone (Table S1). Top: Immunoblotting using anti-mCherry antibody to detect mCherry-CARD8. The full-length (mC-CARD8) and FIIND-processed bands are indicated as well as the HIV^{PR} cut product. The band at ~45 kDa is the result of cleavage by the 20S proteasome (Hsiao et al., 2022). % CARD8 cleavage was calculated by quantifying the HIV^{PR} cut band relative to the HIV-1_{LAI} control using BioRad Image Lab 6. Middle: Immunoblotting with an anti-p24^{gag} antibody showing HIV-1^{gag} cleavage products p41^{gag} and p24^{gag}, and/or full-length HIV-1^{gag}, p55^{gag}. %p24/p55 was calculated from the ratio of p24^{gag} versus p55^{gag} product by quantifying the volume of the p24^{gag} bands versus the p55^{gag} band relative to the HIV-1_{LAI} control using BioRad Image lab 6. Bottom: Immunoblotting with an anti-vinculin antibody to detect vinculin as a loading control. (B) HEK293T cells were transfected with human caspase 1 and human pro-IL-1 β , and either carrier vector or indicated HIV-1 proviruses then probed for IL-1 β secretion 24 hours post-transfection via IL-1R reporter assay. (C) HEK293T cells were transfected with indicated HIV-1 proviruses (300ng). 24 hours post-transfection either wildtype (WT) or *CARD8* knockout (KO) THP-1s were overlaid on the transfected HEK293T cells in a 1:1 ratio. THP-1s were primed with Pam3CSK4 overnight prior to coculture. Supernatants were harvested 24 hours post coculture to assay for IL-1 β secretion as in (B). Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (n=4 biological replicates). p-Values were determined by two-way ANOVA with Dunnett's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

Thus, we reconstituted the inflammasome pathway in HEK293T cells by co-transfection of human caspase 1, human pro-IL-1 β , and either empty vector, HIV-1_{LAI} or representative PI-R proviruses then quantified CASP1-dependent processing of pro-IL-1 β as a readout of CARD8 inflammasome activation as in (Tsu et al., 2023 [DOI](#)). Consistent with the observed differences in CARD8 cleavage by PI-R clones (**Figure 5A** [DOI](#)), we found that PI-R2 and PI-R9, which exhibited less CARD8 cleavage than HIV-1_{LAI}, also induced lower IL-1 β levels than HIV-1_{LAI} (**Figure 5B** [DOI](#)). Similarly, PI-R12 and PI-R13, which demonstrated enhanced CARD8 cleavage, elicited higher IL-1 β levels than HIV-1_{LAI} (**Figure 5B** [DOI](#)). However, these PI-R clones, relative to the LAI strain, may have distinct protease substrate specificity, variable efficiency/kinetics in viral assembly, gag dimerization, and other factors which may also influence CARD8 inflammasome activation. We next assessed inflammasome activation by the PI-R clones in a cell-to-cell transmission model using HEK293T cells as donor cells rather than SUPT1 cells and either WT or CARD8 KO THP-1 cells as the target line at a 1:1 ratio. We opted to overexpress the HIV-1_{LAI} or the PI-R proviruses in HEK293T cells rather than infecting SUPT1 cells due to dramatic variability in replication kinetics between PI-R strains. In these HEK293T:THP-1 cocultures, we observed that cell-to-cell transmission of PI-R2 and PI-R9 resulted in lower IL-1 β levels while PI-R12 and PI-R13 resulted in higher IL-1 β levels compared to HIV-1_{LAI}, respectively (**Figure 5C** [DOI](#)), consistent with our findings from CARD8 cleavage (**Figure 5A** [DOI](#)) and reconstituted inflammasome assays (**Figure 5B** [DOI](#)). Our findings suggest that HIV-dependent inflammasome activation is under genetic control of the viral protease in a manner that can be increased or decreased with naturally occurring mutations induced by drug resistance.

Table S1 [DOI](#) shows the protease inhibitor-resistant (PI-R) clones assayed in **Figure 5** [DOI](#) with corresponding mutations in HIV protease (HIV^{PR}) and HIV^{gag}. ¹These clones were previously cloned and assayed for PI-R in (Varghese et al., 2013 [DOI](#)). The PI-R subset used in **Figure 5B** [DOI](#) are bolded and highlighted in red or green and denote either hypo- or hyper-active CARD8 cleavage, respectively. The last column reports additional amino acid changes in the PI-R clones that were observed via whole plasmid Oxford Nanopore sequencing. *We were unable to sequence verify PI-R3 due to poor plasmid quality. NFV- nelfinavir; FPV- fosamprenavir; SQV- saquinavir; IDV- indinavir; LPV- lopinavir; TPV- tipranavir; DRV- darunavir. The consensus subtype B sequence can be found on the Stanford HIV Drug Resistance Database (HIVDB) ("Stanford - HIV Drug Resistance Database," n.d. [DOI](#)). Relative CARD8 cleavage was determined by quantifying band volume of the CARD8 cleavage product in BioRad Image Lab 6 and comparing to cleavage with HIV-1_{LAI}.

Discussion

We demonstrate that cell-to-cell transmission of HIV-1 from T cells to myeloid cells in both immortalized and primary cell coculture models of infection yields CARD8-dependent inflammasome activation via incoming HIV^{PR}. This inflammasome activation occurs in a largely NLRP3-independent manner. In addition, we identified protease inhibitor resistant strains of HIV-1 that differentially cleave and activate the CARD8 inflammasome. Thus, HIV^{PR} mutants selected for their resistance to different protease inhibitors also affect their ability to cleave host proteins including the inflammasome-forming sensor CARD8.

CARD8 is the primary inflammasome-forming sensor of HIV-1 infection

Previously, both the NLRP3 and IFI16 inflammasomes have been implicated as innate sensors of HIV-1 infection and drivers of CD4⁺ T cell depletion using blood and lymphoid-derived CD4⁺ T cells, respectively, and cell-to-cell transmission was reported to be crucial for IFI16 sensing of abortive HIV transcripts (Doitsh et al., 2014 [DOI](#); Galloway et al., 2015 [DOI](#); Monroe et al., 2014 [DOI](#); Zhang et al., 2021 [DOI](#)). However, the mechanism of NLRP3 inflammasome activation in response to HIV-1 remains elusive. Similarly, there have been reports that IFI16 is not an inflammasome-

Clone name	Reported PI-resistance mutations in HIV ^{PR}	HIV ^{gag} mutation	Strongest PI-R ¹	% CARD8 cleavage (relative to HIV _{LAI})	Additional amino acid changes in HIV ^{PR} relative to NL4.3
HIV-1 _{LAI}	wildtype	wildtype	wildtype	100	37S
CA126802 (PI-R1)	11I, 32I, 33F, 46I, 47V, 54M, 58E, 73S, 84V, 89V, 90M	431V (NC/p1) 437N (NC/p1) 453LF (p1/p6)	FPV, LPV, TPV, DRV	51	10I, 12K, 13V, 20V, 35G, 36I, 37D, 57K, 63P, 64V, 66V, 71V
CA122805 (PI-R2)	10F, 33F, 43T, 46L, 54V, 82A, 84V, 90M	431V (NC/p1) 532S (p1/p6)	SQV	7	16A, 19I, 20R, 35D, 36L, 37D, 55R, 57K, 60E, 62V, 63P, 71V, 93L
CA126805 (PI-R3)	33F, 43T, 46I, 48V, 50V, 54S, 82A	437N (NC/p1) 449VF (p1/p6) PTAP insertion (p6)	SQV, IDV, LPV	8	not determined*
CA96457 (PI-R5)	48V, 53L, 54V, 82A, 90M	436R (NC/p1)	SQV	11	10I, 37D, 63P, 71T, 77I, 93L
CA96458 (PI-R9)	10F, 30N, 33F, 43T, 84V, 88D, 90M	431V (NC/p1)	NFV, SQV	24	15V, 35D, 36L, 37E, 60E, 62V, 63P
CA50834-1 (PI-R10)	24I, 46L, 54V, 76V, 82A	431V (NC/p1)	IDV, LPV	38	10I, 14R, 35D, 36I, 37E, 63P, 71V
CA96451 (PI-R12)	32I, 33F, 43T, 46I, 47V, 54M, 73S, 82A, 89V, 90M	437N (NC/p1) PTAP insertion (p6)	FPV, LPV	192	10V, 12V, 13V, 15V, 20M, 60E, 61N, 62V, 63P, 67Y, 69K, 71I, 72L, 77I
CA20392-1 (PI-R13)	24I, 46L, 54V, 82A	431V (NC/p1)	LPV	206	10I, 14R, 35D, 36I, 37E, 63P, 71V

Table S1

Protease inhibitor resistance mutations and relative CARD8 cleavage.

forming sensor, and instead a nuclear transcriptional regulator of antiviral genes including type I interferons and RIG-I (Hornung et al., 2009 [↗](#); Jiang et al., 2021 [↗](#); Thompson et al., 2014 [↗](#)), suggesting that there may be other mechanisms of CD4⁺ T cell depletion and HIV-dependent inflammasome activation at play. Indeed, CARD8, which is expressed and functional in naïve and memory CD4⁺ and CD8⁺ T cells (Linder et al., 2020 [↗](#)), was recently shown to be required for pyroptosis in primary human blood- and lymphoid-derived CD4⁺ T cells and humanized mouse models (Wang et al., 2024 [↗](#)), implicating CARD8 as a major driver of CD4⁺ T cell depletion during HIV-1 infection. In this study and our prior work (Kulsuptrakul et al., 2023 [↗](#)), we demonstrate that CARD8 is also the primary innate sensor during HIV-1 infection in myeloid cell types during cell-to-cell transmission. However, our present study does not rule out the possibility that under certain conditions or in certain cell types, NLRP3 inflammasome activation may occur, for example following GSDMD pore formation following CARD8 inflammasome activation and play a more profound role in promoting HIV-dependent inflammation. Nonetheless, these data along with other recent work (Wang et al., 2024 [↗](#)) strongly suggest that CARD8 is a major innate sensor of HIV-1 infection.

Protease inhibitor resistance mutations and inflammatory disease

Given the important role of HIV^{PR} in replication, early combination antiretroviral therapy for people living with HIV (PLWH) included protease inhibitors along with RT inhibitors. However, resistance mutations to protease inhibitors quickly arose in PLWH through mutations around the HIV^{PR} active site allowing for polyprotein processing and viral maturation while avoiding drug inhibition. Despite typically having poor overall viral fitness due to less efficient polyprotein processing and replication relative to wildtype HIV-1 in the absence of protease inhibitors, these mutant drug-resistant HIV-1 strains can persist in PLWH on antiviral therapy, posing a major threat to controlling disease progression (De Luca, 2006 [↗](#); Martinez-Picado et al., 1999 [↗](#); Prado et al., 2002 [↗](#); Resch et al., 2002 [↗](#)). To compensate for mutations in HIV^{PR} that change its substrate specificity, HIV^{gag} sometimes evolves mutations around HIV^{PR} cleavage sites to permit proper polyprotein processing (Varghese et al., 2013 [↗](#)). Here, we identified multiple HIV^{PR} inhibitor-resistant strains of HIV-1 that can differentially cleave and activate the CARD8 inflammasome (Figure 5 [↗](#), Table S1 [↗](#)). As the degree of inflammation is a better predictor of disease progression in untreated individuals than viral load (Deeks et al., 2004 [↗](#); Giorgi et al., 1999 [↗](#)), we speculate that differential CARD8 inflammasome activation could influence disease progression for PLWH harboring HIV^{PR} resistance mutations that cleave CARD8 more or less efficiently. More broadly, we suggest that host targets of viral proteases like CARD8 may influence the selection of viral variants during treatment with antiviral protease inhibitor monotherapies.

Viral protease influx activates the CARD8 inflammasome

In this study, we found that HIV-dependent CARD8 inflammasome activation during cell-free infection requires a cationic polymer like DEAE-dextran to facilitate efficient viral infection (see **Supplemental Note**). Despite being infected with the same amount of virus and exhibiting similar percent infection 24 hours post-infection, as measured by intracellular p24^{gag}, with and without DEAE-dextran, we hypothesize that DEAE-dextran during cell-free infection may increase the total viral dose that enters cells consisting of both infectious particles and non-infectious particles that may nonetheless contain active protease, leading to more efficient viral protease influx to trigger CARD8 sensing. Hence, the percentage of p24^{gag} positive cells after 24 hours may be an underestimate of total amount of viral entry in the DEAE-dextran condition. We speculate that a considerable influx of incoming HIV^{PR} may be necessary to induce CARD8 inflammasome activation.

We also observed that DEAE-dextran can trigger inflammasome activation in some THP-1 cell lines (see **Supplemental Note**), prompting us to assay for inflammasome activation upon HIV-1 cell-to-cell transmission with target cells at a 1:1 ratio in the absence of cationic polymer. We found that inflammasome activation following cell-to-cell transmission of HIV-1 could be detected by 24

hours (**Figure 3B**), which is delayed relative to our detection of CARD8 inflammasome activation 2 hours post cell-free HIV-1 infection in the presence of DEAE-dextran (Kulsuptrakul et al., 2023). Nonetheless, we still hypothesize that this inflammasome activation is driven by active incoming viral protease because treatment with an RT inhibitor has no effect on inflammasome activation, which implies that *de novo* protease production is not necessary (**Figure 3A** and **B**, **Figure 4A** and **B**). Similarly, we observed that IL-1 β levels do not increase after plateauing 24 hours after establishing the coculture (**Figure 3B**, **Figure 4B**), suggesting that secondary infection does not further amplify inflammasome activation. We infer that this is also likely a product of the efficiency of viral entry and the necessity for multiple virions infecting at the same time to deliver a sufficient amount of active HIV^{PR} for cytosolic CARD8 sensing. We postulate that under certain physiological conditions, cell-to-cell transmission can cause CARD8 inflammasome activation when there is an influx of active incoming HIV^{PR} across the viral synapse. Taken together, we speculate that both cell-free infection facilitated by cationic polymer or cell-to-cell transmission can achieve sufficient levels of active HIV^{PR} influx to activate the CARD8 inflammasome.

Macrophages have been reported to be primarily infected through phagocytosis of infected CD4⁺ T cells or cell-to-cell transmission (Dupont and Sattentau, 2020; Martínez-Méndez et al., 2017; Orenstein, 2000). We demonstrate that unlike CD4⁺ T cells, which are rapidly depleted by HIV-1 infection and do not release IL-1 β or IL-18 (Linder et al., 2020), primary macrophages release pro-inflammatory cytokines in response to HIV^{PR} during cell-to-cell infection from primary CD4⁺ T cells and T cell lines (**Figure 3**, **Figure 4**), thus representing a potential source of sustained IL-1 β and subsequent chronic immune activation. In addition to promoting chronic immune activation, HIV-dependent IL-1 β release from macrophages may also contribute to HIV-1 pathogenesis by activating nearby CD4⁺ T cells, rendering them susceptible to becoming infected with HIV-1, and thus indirectly promoting CD4⁺ T cell depletion. Collectively with our prior work (Kulsuptrakul et al., 2023), our findings provide further evidence that CARD8 inflammasome activation is driven by incoming HIV^{PR} under conditions where multiple virions may enter cells, and thus could be a potential driver of HIV-1 pathogenesis by promoting chronic immune activation.

Methods

Plasmids and Reagents

pMD2.G used for HIV-1_{LAI-VSVG} production was a gift from Didier Trono (Addgene). HIV-1_{LAI} has been previously described (Peden et al., 1991). The following reagents were obtained through the NIH HIV Reagent Program, Division of AIDS, NIAID, NIH: lopinavir (LPV), nevirapine (NVP), Human Immunodeficiency Virus 1 (HIV-1) NL4-3 BaL Infectious Molecular Clone (p20-36) (HIV-1_{NL4.3-BaL}), ARP-11442, contributed by Dr. Bruce Chesebro (Chesebro et al., 1992, 1991; Toohey et al., 1995; Walter et al., 2005), and Panel of Multi-Protease Inhibitor Resistant Infectious Molecular Clones, HRP-12740, contributed by Dr. Robert Shafer (Varghese et al., 2013). Mutant HIV^{PR} sequences were amplified from clinically-derived viral cDNA encoding protease genes with resistance to multiple PRIs then cloned into an NL4.3 backbone with overhangs including the 3' end of gag with the gag cleavage site and the 5' end of RT as previously described (Varghese et al., 2013). CARD8 variant constructs were cloned as previously described (Kulsuptrakul et al., 2023). VX765 and MCC950 were sourced from Invivogen (cat: inh-vx765i-1 and inh-mcc, respectively).

Cell culture

SUPT1 (ATCC) and THP-1 cells (JK and ATCC) were cultured in RPMI (Invitrogen) with 10% FBS, 1% penicillin/streptomycin antibiotics, 10 mM HEPES, 0.11 g/L sodium pyruvate, 4.5 g/L D-glucose and 1% Glutamax. JK THP-1 cells were used for all experiments in this manuscript and our previous

work unless explicitly stated (see **Supplemental Note**) ([Kulsuptrakul et al., 2023](#)). Primary monocytes were cultured in RPMI (Invitrogen) with 10% FBS, and 1% penicillin/streptomycin antibiotics and differentiated in the presence of 20ng/mL GM-CSF (Peprotech cat: 300-03) and 20ng/mL M-CSF (Peprotech cat: 300-25). Primary CD4⁺ T cells were cultured in RPMI (Invitrogen) with 10% FBS, 1% penicillin/streptomycin antibiotics and 100U/mL IL-2. HEK293T (ATCC) lines were cultured in DMEM (Invitrogen) with 10% FBS and 1% penicillin/streptomycin antibiotics. All lines routinely tested negative for mycoplasma bacteria (Fred Hutch Specimen Processing & Research Cell Bank). SUPT1 and THP-1 cell lines were authenticated by STR profiling analysis (Fred Hutch Genomics core and TransnetYX, Inc.) (see **Supplemental Note**).

HIV-1_{LAI}, HIV-1_{LAI-VSVG}, and HIV-1_{NL4.3-BaL} production

293T cells were seeded at $2\text{--}3 \times 10^5$ cells/well in six-well plates the day before transfection using TransIT-LT1 reagent (Mirus Bio LLC) at 3 μL transfection reagent/well as previously described ([OhAinle et al., 2018](#)). For HIV-1 production, 293Ts were transfected with 1 μg /well HIV_{LAI} or HIV-1_{NL4.3-BaL} proviral DNA or 1 μg /well HIV_{LAI} Δenv DNA and 500 ng/well pMD2.G for HIV-1_{LAI}, HIV-1_{NL4.3-BaL}, and HIV-1_{LAI-VSVG}, respectively. One day post-transfection, media was replaced. Two days post-transfection, viral supernatants were collected and filtered through a 20 μm filter and aliquots were frozen at -80°C . HIV-1_{LAI}, HIV-1_{NL4.3-BaL} and HIV-1_{LAI-VSVG} proviruses were previously described ([Bartz and Vodicka, 1997](#); [Gummuluru et al., 2003](#); [Peden et al., 1991](#)).

Cell-free and cell-to-cell coculture HIV-1 infection

Cell-free infections with HIV-1_{LAI-VSVG} were done as previously described ([Kulsuptrakul et al., 2023](#)). Subsequent cell death was assessed by incubating in media containing propidium iodide dye (10 μg /mL) for 5 minutes at room temperature then washed once with PBS before fixing with BD CytoFix/Cytoperm (cat:BDB554714) and staining for intracellular p24^{gag} (Beckman Coulter cat#: 6604665) for flow cytometry. In the HIV-1 cell-to-cell transmission system, SUPT1 expressing CCR5 (SUPT1-CCR5) cells were spinoculated at 1100g for 30min with either HIV-1_{LAI} or HIV-1_{NL4.3-BaL} in the presence of 10 μg /mL DEAE-dextran. SUPT1-CCR5 cells were lentiviral transduced to express CCR5 ([Dingens et al., 2017](#)). After 24 hours, mock or HIV-1 infected SUPT1-CCR5 cells were washed three times in PBS such that DEAE-dextran and cell-free virus were removed before starting coculture with THP-1 cells or MDMs. THP-1 cells and MDMs were seeded at 5×10^5 cells/well and primed with 500ng/mL Pam3CSK4 (Invivogen) for 16-24 hours before coculture. Mock or infected SUPT1 cells were seeded at 5×10^5 cells/well. Cultured supernatants from coculture were harvested 48 hours after starting the coculture for the IL-1R reporter assay, which was previously described ([Kulsuptrakul et al., 2023](#)).

Transwell coculture HIV-1 infection

SUPT1 cells were spinoculated at 1100g for 30min with HIV-1_{LAI} in the presence of 10 μg /mL DEAE-dextran. After 24 hours, mock or HIV-1 infected SUPT1 cells were washed 3 times in PBS and either mixed in a 24-well with THP-1 cells or placed in a transwell insert above target THP-1 cells at a concentration of 5×10^5 infected SUPT1 cells and 2.5×10^5 THP-1 cells per well. THP-1 cells were primed overnight with 500ng/mL Pam3CSK4 before starting coculture. The transwell insert has a 0.4 μm membrane at the bottom of the well (ThinCert™ Tissue Culture Inserts, Sterile, Greiner Bio-One cat:665640), allowing virus to pass out of the transwell but not the infected cell. Reverse transcriptase (RT) activity in viral supernatants was measured using the RT activity assay as previously described ([Roesch et al., 2018](#); [Vermeire et al., 2012](#)). A stock of HIV-1_{LAI} virus was titrated multiple times, aliquoted at -80°C and used as the standard curve in all assays.

Monocyte-derived macrophage isolation, differentiation, and editing

Primary monocytes were isolated via negative selection using the EasySep™ Human Monocyte Isolation Kit (Easy Sep, 1×10^9) (Stem Cell Technologies) according to the manufacturer's protocols from PBMCs collected from blood donors. Upon isolation, monocytes were seeded at 1×10^6 cell/mL and differentiated for 5 days in the presence of media containing 20ng/mL GM-CSF (PeproTech cat: 300-03) and 20ng/mL M-CSF (PeproTech cat: 300-25), changing media every other day. For edited MDMs, isolated monocytes were electroporated in cuvettes (100µL) with $2.5\text{--}5 \times 10^6$ cells/nucleofection in the presence of pre-complexed Cas9-RNPs (300pmol sgRNA: 100pmol Cas9) in Lonza P2 buffer using pulse code DK-100. RNPs were complexed with sgRNA from the Synthego gene KO kit, which includes 3 sgRNAs per gene. Thus, each sgRNA was present at a 1:1 ratio with Cas9 (QB3 MacroLab or Synthego SpCas9 2NLS Nuclease). A table of sgRNAs used for *AAVS1*, *CARD8*, or *NLRP3* KO can be found in [Table S2](#) below. After nucleofection, cells were supplemented with 900µL of prewarmed media and allowed to recover for 30 minutes at 37°C before counting and seeding at $\sim 1\text{--}1.5 \times 10^6$ cells/mL for differentiation. Media was changed 24 hours post nucleofection then differentiated for 5 more days before characterizing knockout efficiency and conducting coculture experiments.

CD4+ T cell isolation, infection, and coculture

Primary CD4+ T cells were isolated via positive selection using the EasySep™ Release Human CD4 Positive selection kit (Stem Cell Technologies Cat: 17752) according to the manufacturer's instructions from PBMCs collected from blood donors and seeded at 2.5×10^6 cells/mL in the presence of 100U/mL IL-2. T cells were activated 24 hours post-isolation with CD3/CD28 activation beads (Miltenyi Biotech Cat: 130-091-441). Activation beads were removed according to the manufacturer's protocols 24 hours later for infection. For infection, T cells were suspended at $1\text{--}1.5 \times 10^6$ cells/mL in 15mL conical tubes containing 8µg/mL polybrene, 100U/mL IL-2, and HIV-1_{NL4.3-BaL} then spinoculated at 1100g at 30°C for 90 min. Three days post-infection, CD4s were assessed for intracellular p24^{gag} via flow cytometry ($\sim 10\%$ infected) then washed thrice with PBS before coculturing with MDMs. CD4:MDM coculture RPMI media was supplemented with 100U/mL IL-2, 20ng/mL GM-CSF, 20ng/mL M-CSF, and 500ng/mL Pam3CSK4.

CARD8 cleavage assay

HEK293T cells were seeded at $1\text{--}1.5 \times 10^5$ cells/well in 24-well plates the day before transfection using TransIT-LT1 reagent at 1.5 µL transfection reagent/well (Mirus Bio LLC). One hundred ng of indicated constructs encoding an N-terminal mCherry-tagged CARD8 were co-transfected into HEK293T cells with empty vector (‘-’), HIV_{LAI} or PI-R provirus. To normalize HIV^{gag} expression between HIV-1_{LAI} and the PI-R clones, which are in a different vector backbone, 400ng of HIV-1_{LAI} and 200ng of all PI-R clones were transfected. All conditions were normalized with empty vector to contain the same amount of DNA. Cytoplasmic lysates were harvested 24 hours post-transfection and immunoblotted as previously described ([Kulsuptrakul et al., 2023](#)).

HEK reconstitution assay

HEK293T cells, which endogenously express CARD8, were seeded at 2.25×10^5 cell/well in 24-well plates the day before transfection using TransIT-LT1 reagent at 1.5uL transfection reagent/well (Mirus Bio LLC). Functional inflammasomes were reconstituted by transfecting in 5ng human CASP1 and 100ng human IL-1β. To assess the effects of different viral proteases on inflammasome activation, HIV-1_{LAI} or PI-R clones were co-transfected in with CASP1 and IL-1β. As with the CARD8 cleavage assay, a higher amount of 250ng HIV-1_{LAI} was added relative to the PI-R clones, which were all added at 125ng, to normalize HIV^{gag} expression between the different vector backbones.

sgRNA	Sequence
CARD8 sgRNA1	CUCUGCAGUGACAUCAAACA
CARD8 sgRNA2	UGACGAUUGCGUUUGGUUCC
CARD8 sgRNA3	AGCGUUUGGUUCCCCACUGC
AAVS1 sgRNA 1	GUUAAUGUGGCUCUGGUUCU
AAVS1 sgRNA 2	ACCCACAGUGGGGCCACUA
AAVS1 sgRNA 3	CCUCCUAGUCUCCUGAUAU
NLRP3 sgRNA 1	GCUCAGAAUGCUCAUCAUCG
NLRP3 sgRNA 2	GAUGAUGUUGGACUGGGCAU
NLRP3 sgRNA 3	CAAGGCUCACCUCCCGACAG

Table S2

All conditions were normalized with empty vector to contain the same amount of DNA. Cultured supernatant was harvested 24 hours post-transfection to assay for IL-1 β secretion via IL-1R reporter assay.

Supplemental Note

There have been 4 different sublines of THP-1 cells previously characterized (Kasai et al., 2022 [link](#)). Using short tandem repeat (STR) profiling, we were able to distinguish the WT THP-1 cell stocks that were used as the parental line for knockouts and complemented knockouts in this work and our prior work (Kulsuptrakul et al., 2023 [link](#)), as distinct from WT THP-1 cells sourced from ATCC at 3 different loci (**Figure A1A** [link](#)). Of note, unlike the THP-1 cells used here (referred to as JK THP-1), ATCC THP-1 cells elicited IL-1 β secretion in the absence of HIV-1 infection in the presence of 20 μ g/mL DEAE-dextran (**Figure A1B** [link](#)). Unless otherwise specified, any mention of “THP-1 cells” are referring to our JK THP-1 cells, not ATCC THP-1 cells. Nonetheless, given the sensitivity of some THP-1 sublines to elicit an inflammasome response in the presence of DEAE-dextran, we assessed whether or not we could establish systems to measure HIV-1 induced CARD8-dependent inflammasome activation in the absence of DEAE-dextran. Thus, we infected either wildtype (WT) or *CARD8* knockout (KO) THP-1 cells with wildtype HIV-1_{LAI} in either the presence or absence of DEAE-dextran and measured cell death and IL-1 β secretion 24 hours post-infection as readouts of inflammasome activation. We found that despite achieving similar levels of infection (20-30%) as measured by intracellular p24^{gag} after spinoculation with and without DEAE-dextran (**Figure A1C** [link](#), left), we detected robust CARD8-dependent inflammasome activation in WT THP-1 cells infected only in the presence of DEAE-dextran (**Figure A1C** [link](#), middle and right). These data suggest that cationic polymer is necessary to observe HIV-dependent CARD8 inflammasome activation in our cell-free system.

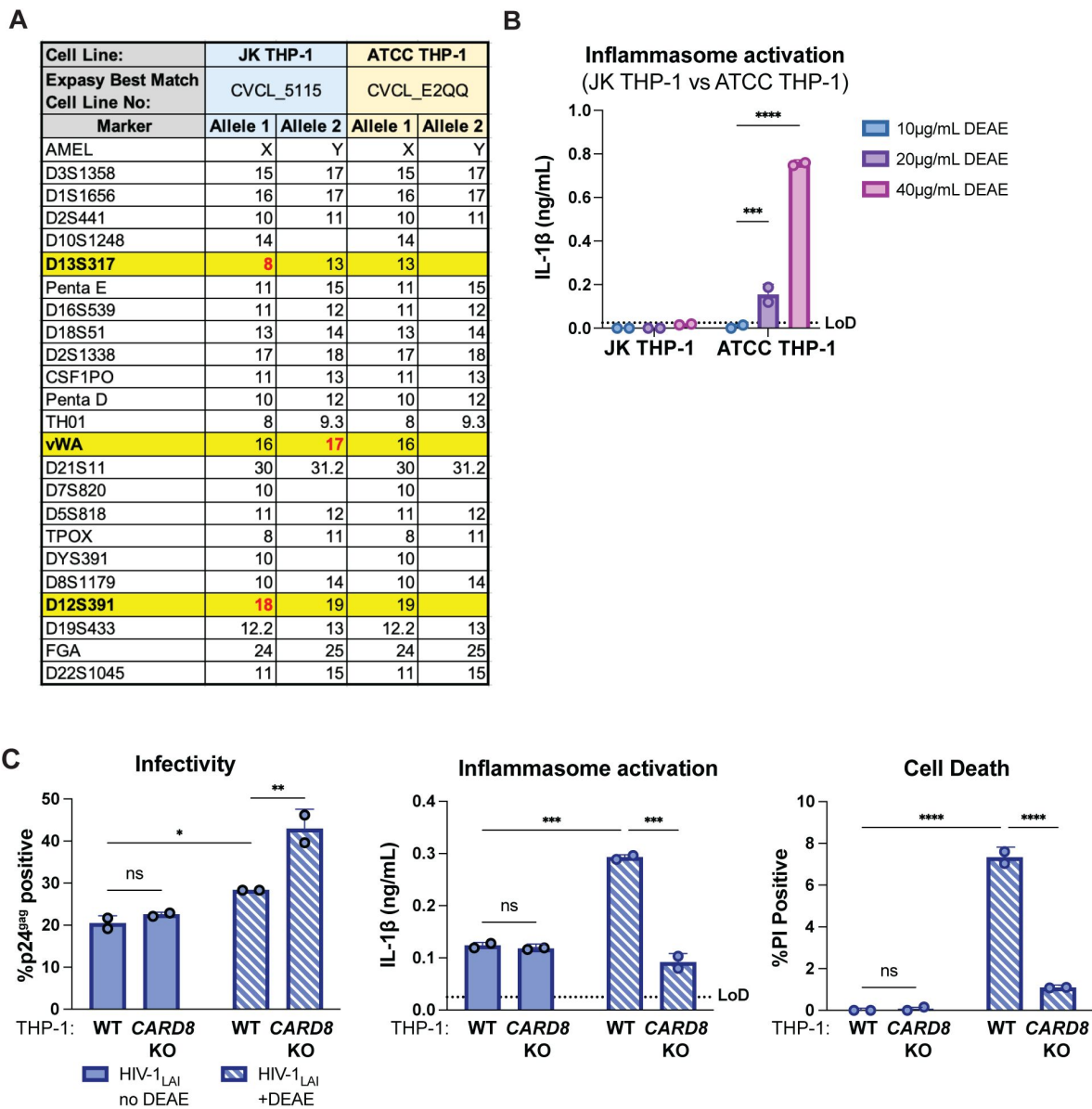


Figure A1.

Characterization of THP-1 cells.

(A) Promega GenePrint® 24 system STR analysis summary of our JK THP-1 cells versus ATCC THP-1 cells. Cell line authentication was done by TransnetYX, Inc. by following the protocol described in ANSI/ATCC ASN-0002-2011. The STR alleles were searched on the ATCC Database and the Expaty best match cell numbers for each cell line had a 100% database match. Distinguishing loci are highlighted in yellow and distinguishing alleles are in red. (B) JK and ATCC THP-1 cells were primed with Pam3CSK4 overnight then treated with increasing doses of DEAE-dextran for 24 hours before probing for IL-1 β secretion. (C) Wildtype (WT) or *CARD8* knockout (KO) THP-1 cells were infected with wildtype HIV-1_{LAI} at the same MOI in the presence or absence of DEAE-dextran (10 μ g/mL) then harvested after 24 hours and assayed for: left) percent infection via intracellular p24^{tag}, middle) inflammasome activation by IL-1 β secretion via IL-1R reporter assay, and right) cell death via propidium iodide (PI) dye uptake using flow cytometry. %PI positive and IL-1 levels are normalized to mock control. Dotted line indicates limit of detection (LoD). Datasets represent mean $\pm$ SD (n=2 biological replicates). Two-way ANOVA with (B) Sidak's or (C) Tukey's test using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

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References

- Abela IA, Berlinger L, Schanz M, Reynell L, Günthard HF, Rusert P, Trkola A (2012) **Cell-Cell Transmission Enables HIV-1 to Evade Inhibition by Potent CD4bs Directed Antibodies** *PLOS Pathog* **8**:e1002634 <https://doi.org/10.1371/journal.ppat.1002634>
- Agosto LM, Herring MB, Mothes W, Henderson AJ (2018) **HIV-1-Infected CD4+ T Cells Facilitate Latent Infection of Resting CD4+ T Cells through Cell-Cell Contact** *Cell Rep* **24**:2088–2100 <https://doi.org/10.1016/j.celrep.2018.07.079>
- Agosto LM, Uchil PD, Mothes W (2015) **HIV cell-to-cell transmission: effects on pathogenesis and antiretroviral therapy** *Trends Microbiol* **23**:289–295 <https://doi.org/10.1016/j.tim.2015.02.003>
- Bailey CA, Miller DK, Lenard J (1984) **Effects of DEAE-dextran on infection and hemolysis by VSV. Evidence that nonspecific electrostatic interactions mediate effective binding of VSV to cells** *Virology* **133**:111–118 [https://doi.org/10.1016/0042-6822\(84\)90429-x](https://doi.org/10.1016/0042-6822(84)90429-x)
- Bandera A, Masetti M, Fabbiani M, Biasin M, Muscatello A, Squillace N, Clerici M, Gori A, Trabattoni D (2018) **The NLRP3 Inflammasome Is Upregulated in HIV-Infected Antiretroviral Therapy-Treated Individuals with Defective Immune Recovery** *Front Immunol* **9**
- Bartz SR, Vodicka MA (1997) **Production of High-Titer Human Immunodeficiency Virus Type 1 Pseudotyped with Vesicular Stomatitis Virus Glycoprotein** *Methods* **12**:337–342 <https://doi.org/10.1006/meth.1997.0487>
- Baxter AE, Russell RA, Duncan CJA, Moore MD, Willberg CB, Pablos JL, Finzi A, Kaufmann DE, Ochsenbauer C, Kappes JC, Groot F, Sattentau QJ (2014) **Macrophage infection via selective capture of HIV-1-infected CD4+ T cells** *Cell Host Microbe* **16**:711–721 <https://doi.org/10.1016/j.chom.2014.10.010>
- Brenchley JM, Paiardini M, Knox KS, Asher AI, Cervasi B, Asher TE, Scheinberg P, Price DA, Hage CA, Kholi LM, Khoruts A, Frank I, Else J, Schacker T, Silvestri G, Douek DC (2008) **Differential Th17 CD4 T-cell depletion in pathogenic and nonpathogenic lentiviral infections** *Blood* **112**:2826–2835 <https://doi.org/10.1182/blood-2008-05-159301>
- Brenchley JM, Price DA, Schacker TW, Asher TE, Silvestri G, Rao S, Kazzaz Z, Bornstein E, Lambotte O, Altmann D, Blazar BR, Rodriguez B, Teixeira-Johnson L, Landay A, Martin JN, Hecht FM, Picker LJ, Lederman MM, Deeks SG, Douek DC (2006) **Microbial translocation is a cause of systemic immune activation in chronic HIV infection** *Nat Med* **12**:1365–1371 <https://doi.org/10.1038/nm1511>
- Broz P, Dixit VM (2016) **Inflammasomes: mechanism of assembly, regulation and signalling** *Nat Rev Immunol* **16**:407–420 <https://doi.org/10.1038/nri.2016.58>
- Castro LK, Daugherty MD (2023) **Tripping the wire: sensing of viral protease activity by CARD8 and NLRP1 inflammasomes** *Curr Opin Immunol* **83**:102354 <https://doi.org/10.1016/j.coi.2023.102354>

- Chen P, Hübner W, Spinelli MA, Chen BK (2007) **Predominant mode of human immunodeficiency virus transfer between T cells is mediated by sustained Env-dependent neutralization-resistant virological synapses** *J Virol* **81**:12582–12595 <https://doi.org/10.1128/JVI.00381-07>
- Chesebro B, Nishio J, Perryman S, Cann A, O'Brien W, Chen IS, Wehrly K (1991) **Identification of human immunodeficiency virus envelope gene sequences influencing viral entry into CD4-positive HeLa cells, T-leukemia cells, and macrophages** *J Virol* **65**:5782–5789 <https://doi.org/10.1128/JVI.65.11.5782-5789.1991>
- Chesebro B, Wehrly K, Nishio J, Perryman S (1992) **Macrophage-tropic human immunodeficiency virus isolates from different patients exhibit unusual V3 envelope sequence homogeneity in comparison with T-cell-tropic isolates: definition of critical amino acids involved in cell tropism** *J Virol* **66**:6547–6554 <https://doi.org/10.1128/JVI.66.11.6547-6554.1992>
- Chivero ET, Guo M-L, Periyasamy P, Liao K, Callen SE, Buch S (2017) **HIV-1 Tat Primes and Activates Microglial NLRP3 Inflammasome-Mediated Neuroinflammation** *J Neurosci* **37**:3599–3609 <https://doi.org/10.1523/JNEUROSCI.3045-16.2017>
- Chun T-W, Carruth L, Finzi D, Shen X, DiGiuseppe JA, Taylor H, Hermankova M, Chadwick K, Margolick J, Quinn TC, Kuo Y-H, Brookmeyer R, Zeiger MA, Barditch-Crovo P, Siliciano RF (1997a) **Quantification of latent tissue reservoirs and total body viral load in HIV-1 infection** *Nature* **387**:183–188 <https://doi.org/10.1038/387183a0>
- Chun T-W, Finzi D, Margolick J, Chadwick K, Schwartz D, Siliciano RF (1995) **In vivo fate of HIV-1-infected T cells: Quantitative analysis of the transition to stable latency** *Nat Med* **1**:1284–1290 <https://doi.org/10.1038/nm1295-1284>
- Chun T-W, Stuyver L, Mizell SB, Ehler LA, Mican JAM, Baseler M, Lloyd AL, Nowak MA, Fauci AS (1997b) **Presence of an inducible HIV-1 latent reservoir during highly active antiretroviral therapy** *Proc Natl Acad Sci* **94**:13193–13197 <https://doi.org/10.1073/pnas.94.24.13193>
- Clark KM, Kim JG, Wang Q, Gao H, Presti RM, Shan L (2022) **Chemical inhibition of DPP9 sensitizes the CARD8 inflammasome in HIV-1-infected cells** *Nat Chem Biol* **18**:1–9 <https://doi.org/10.1038/s41589-022-01182-5>
- Coll RC, Robertson AAB, Chae JJ, Higgins SC, Muñoz-Planillo R, Inserra MC, Vetter I, Dungan LS, Monks BG, Stutz A, Croker DE, Butler MS, Haneklaus M, Sutton CE, Núñez G, Latz E, Kastner DL, Mills KHG, Masters SL, Schroder K, Cooper MA, O'Neill LAJ (2015) **A small-molecule inhibitor of the NLRP3 inflammasome for the treatment of inflammatory diseases** *Nat Med* **21**:248–255 <https://doi.org/10.1038/nm.3806>
- Collman R, Godfrey B, Cutilli J, Rhodes A, Hassan NF, Sweet R, Douglas SD, Friedman H, Nathanson N, Gonzalez-Scarano F (1990) **Macrophage-tropic strains of human immunodeficiency virus type 1 utilize the CD4 receptor** *J Virol* **64**:4468–4476 <https://doi.org/10.1128/jvi.64.9.4468-4476.1990>
- Collman R, Hassan NF, Walker R, Godfrey B, Cutilli J, Hastings JC, Friedman H, Douglas SD, Nathanson N (1989) **Infection of monocyte-derived macrophages with human immunodeficiency virus type 1 (HIV-1). Monocyte-tropic and lymphocyte-tropic strains of**

HIV-1 show distinctive patterns of replication in a panel of cell types *J Exp Med* **170**:1149–1163 <https://doi.org/10.1084/jem.170.4.1149>

Conant D, Hsiao T, Rossi N, Oki J, Maures T, Waite K, Yang J, Joshi S, Kelso R, Holden K, Enzmann BL, Stoner R (2022) **Inference of CRISPR Edits from Sanger Trace Data** *Crispr J* **5**:123–130 <https://doi.org/10.1089/crispr.2021.0113>

De Luca A (2006) **The impact of resistance on viral fitness and its clinical implications** In: Geretti AM, editors. *Antiretroviral Resistance in Clinical Practice* London: Mediscript

Deeks SG, Kitchen CMR, Liu L, Guo H, Gascon R, Narváez AB, Hunt P, Martin JN, Kahn JO, Levy J, McGrath MS, Hecht FM (2004) **Immune activation set point during early HIV infection predicts subsequent CD4+ T-cell changes independent of viral load** *Blood* **104**:942–947 <https://doi.org/10.1182/blood-2003-09-3333>

Del Portillo A, Tripodi J, Najfeld V, Wodarz D, Levy DN, Chen BK (2011) **Multiploid Inheritance of HIV-1 during Cell-to-Cell Infection** *J Virol* **85**:7169–7176 <https://doi.org/10.1128/JVI.00231-11>

Dingens AS, Haddox HK, Overbaugh J, Bloom JD (2017) **Comprehensive mapping of HIV-1 escape from a broadly neutralizing antibody** *Cell Host Microbe* **21**:777–787 <https://doi.org/10.1016/j.chom.2017.05.003>

Doitsh G, Galloway NL, Geng X, Yang Z, Monroe KM, Zepeda O, Hunt PW, Hatano H, Sowinski S, Muñoz-Arias I, Greene WC (2014) **Pyroptosis drives CD4 T-cell depletion in HIV-1 infection** *Nature* **505**:509–514 <https://doi.org/10.1038/nature12940>

Dufloo J, Bruel T, Schwartz O (2018) **HIV-1 cell-to-cell transmission and broadly neutralizing antibodies** *Retrovirology* **15**:51 <https://doi.org/10.1186/s12977-018-0434-1>

Duncan CJA, Russell RA, Sattentau QJ (2013) **High multiplicity HIV-1 cell-to-cell transmission from macrophages to CD4+ T cells limits antiretroviral efficacy** *Aids* **27**:2201 <https://doi.org/10.1097/QAD.0b013e3283632ec4>

Dupont M, Sattentau QJ (2020) **Macrophage Cell-Cell Interactions Promoting HIV-1 Infection** *Viruses* **12**:492 <https://doi.org/10.3390/v12050492>

Fink SL, Cookson BT (2005) **Apoptosis, Pyroptosis, and Necrosis: Mechanistic Description of Dead and Dying Eukaryotic Cells** *Infect Immun* **73**:1907–1916 <https://doi.org/10.1128/IAI.73.4.1907-1916.2005>

Galloway NLK, Doitsh G, Monroe KM, Yang Z, Muñoz-Arias I, Levy DN, Greene WC (2015) **Cell-to-Cell Transmission of HIV-1 Is Required to Trigger Pyroptotic Death of Lymphoid-Tissue-Derived CD4 T Cells** *Cell Rep* **12**:1555–1563 <https://doi.org/10.1016/j.celrep.2015.08.011>

Giorgi JV, Hultin LE, McKeating JA, Johnson TD, Owens B, Jacobson LP, Shih R, Lewis J, Wiley DJ, Phair JP, Wolinsky SM, Detels R (1999) **Shorter Survival in Advanced Human Immunodeficiency Virus Type 1 Infection Is More Closely Associated with T Lymphocyte Activation than with Plasma Virus Burden or Virus Chemokine Coreceptor Usage** *J Infect Dis* **179**:859–870 <https://doi.org/10.1086/314660>

Gosselin A, Monteiro P, Chomont N, Diaz-Griffero F, Said EA, Fonseca S, Wacleche V, El-Far M, Boulassel M-R, Routy J-P, Sekaly R-P, Ancuta P (2009) **Peripheral Blood CCR4+CCR6+ and**

CXCR3+CCR6+ CD4+ T Cells Are Highly Permissive to HIV-1 Infection *J Immunol* **184**:1604–1616 <https://doi.org/10.4049/jimmunol.0903058>

Gummuluru S, Rogel M, Stamatatos L, Emerman M (2003) **Binding of Human Immunodeficiency Virus Type 1 to Immature Dendritic Cells Can Occur Independently of DC-SIGN and Mannose Binding C-Type Lectin Receptors via a Cholesterol-Dependent Pathway** *J Virol* **77**:12865–12874 <https://doi.org/10.1128/JVI.77.23.12865-12874.2003>

Hernandez JC, Latz E, Urcuqui-Inchima S (2013) **HIV-1 Induces the First Signal to Activate the NLRP3 Inflammasome in Monocyte-Derived Macrophages** *Intervirology* **57**:36–42 <https://doi.org/10.1159/000353902>

Hornung V, Ablasser A, Charrel-Dennis M, Bauernfeind F, Horvath G, Caffrey DR, Latz E, Fitzgerald KA (2009) **AIM2 recognizes cytosolic dsDNA and forms a caspase-1 activating inflammasome with ASC** *Nature* **458**:514–518 <https://doi.org/10.1038/nature07725>

Hsiao JC, Neugroschl AR, Chui AJ, Taabazuing CY, Griswold AR, Wang Q, Huang H-C, Orth-He EL, Ball DP, Hiotis G, Bachovchin DA (2022) **A ubiquitin-independent proteasome pathway controls activation of the CARD8 inflammasome** *J Biol Chem* **298**:102032 <https://doi.org/10.1016/j.jbc.2022.102032>

Iwami S, Takeuchi JS, Nakaoka S, Mammano F, Clavel F, Inaba H, Kobayashi T, Misawa N, Aihara K, Koyanagi Y, Sato K (2015) **Cell-to-cell infection by HIV contributes over half of virus infection** *eLife* **4**:e08150 <https://doi.org/10.7554/eLife.08150>

Jiang Z, Wei F, Zhang Y, Wang T, Gao W, Yu S, Sun H, Pu J, Sun Y, Wang M, Tong Q, Gao C, Chang K-C, Liu J (2021) **IFI16 directly senses viral RNA and enhances RIG-I transcription and activation to restrict influenza virus infection** *Nat Microbiol* **6**:932–945 <https://doi.org/10.1038/s41564-021-00907-x>

Jolly C, Booth NJ, Neil SJD (2010) **Cell-Cell Spread of Human Immunodeficiency Virus Type 1 Overcomes Tetherin/BST-2-Mediated Restriction in T cells** *J Virol* **84**:12185–12199 <https://doi.org/10.1128/jvi.01447-10>

Kulsuptrakul J, Turcotte EA, Emerman M, Mitchell PS (2023) **A human-specific motif facilitates CARD8 inflammasome activation after HIV-1 infection** *eLife* **12**:e84108 <https://doi.org/10.7554/eLife.84108>

Leal VNC, Reis EC, Fernandes FP, Pontillo A (2020) **NLRP3 inflammasome activation profile in CD4+ T and CD19+ B lymphocytes from HIV-infected patients** *J Immunol* **204**:225–8 <https://doi.org/10.4049/jimmunol.204.Supp.225.8>

Linder A, Bauernfried S, Cheng Y, Albanese M, Jung C, Keppler OT, Hornung V (2020) **CARD8 inflammasome activation triggers pyroptosis in human T cells** *EMBO J* **39**:e105071 <https://doi.org/10.15252/emboj.2020105071>

Lopez P, Koh WH, Hnatiuk R, Murooka TT (2019) **HIV Infection Stabilizes Macrophage-T Cell Interactions To Promote Cell-Cell HIV Spread** *J Virol* **93**:e00805–19 <https://doi.org/10.1128/JVI.00805-19>

Mamik MK, Hui E, Branton WG, McKenzie BA, Chisholm J, Cohen EA, Power C (2017) **HIV-1 Viral Protein R Activates NLRP3 Inflammasome in Microglia: implications for HIV-1 Associated Neuroinflammation** *J Neuroimmune Pharmacol* **12**:233–248 <https://doi.org/10.1007/s11481-016-9708-3>

- Martin N, Welsch S, Jolly C, Briggs JAG, Vaux D, Sattentau QJ (2010) **Virological Synapse-Mediated Spread of Human Immunodeficiency Virus Type 1 between T Cells Is Sensitive to Entry Inhibition** *J Virol* **84**:3516–3527 <https://doi.org/10.1128/jvi.02651-09>
- Martínez-Méndez D, Rivera-Toledo E, Ortega E, Licona-Limón I, Huerta L (2017) **Monocyte-lymphocyte fusion induced by the HIV-1 envelope generates functional heterokaryons with an activated monocyte-like phenotype** *Exp Cell Res* **352**:9–19 <https://doi.org/10.1016/j.yexcr.2017.01.014>
- Martinez-Picado J, Savara AV, Sutton L, D'Aquila RT (1999) **Replicative fitness of protease inhibitor-resistant mutants of human immunodeficiency virus type 1** *J Virol* **73**:3744–3752 <https://doi.org/10.1128/JVI.73.5.3744-3752.1999>
- Monroe KM, Yang Z, Johnson JR, Geng X, Doitsh G, Krogan NJ, Greene WC (2014) **IFI16 DNA Sensor Is Required for Death of Lymphoid CD4 T-cells Abortively Infected with HIV** *Science* **343**:428–432 <https://doi.org/10.1126/science.1243640>
- Nadkarni R, Chu WC, Lee CQE, Mohamud Y, Yap L, Toh GA, Beh S, Lim R, Fan YM, Zhang YL, Robinson K, Tryggvason K, Luo H, Zhong F, Ho L (2022) **Viral proteases activate the CARD8 inflammasome in the human cardiovascular system** *J Exp Med* **219**:e20212117 <https://doi.org/10.1084/jem.20212117>
- OhAinle M, Helms L, Vermeire J, Roesch F, Humes D, Basom R, Delrow JJ, Overbaugh J, Emerman M (2018) **A virus-packageable CRISPR screen identifies host factors mediating interferon inhibition of HIV** *eLife* **7**:e39823 <https://doi.org/10.7554/eLife.39823>
- Orenstein JM (2000) **In Vivo Cytolysis and Fusion of Human Immunodeficiency Virus Type 1-Infected Lymphocytes in Lymphoid Tissue** *J Infect Dis* **182**:338–342 <https://doi.org/10.1086/315640>
- Peden K, Emerman M, Montagnier L (1991) **Changes in growth properties on passage in tissue culture of viruses derived from infectious molecular clones of HIV-1LAI, HIV-1MAL, and HIV-1ELI** *Virology* **185**:661–672 [https://doi.org/10.1016/0042-6822\(91\)90537-L](https://doi.org/10.1016/0042-6822(91)90537-L)
- Prado JG, Wrin T, Beauchaine J, Ruiz L, Petropoulos CJ, Frost SDW, Clotet B, D'Aquila RT, Martinez-Picado J (2002) **Amprenavir-resistant HIV-1 exhibits lopinavir cross-resistance and reduced replication capacity** *AIDS Lond Engl* **16**:1009–1017 <https://doi.org/10.1097/00002030-200205030-00007>
- Primiano MJ, Lefker BA, Bowman MR, Bree AG, Hubeau C, Bonin PD, Mangan M, Dower K, Monks BG, Cushing L, Wang S, Guzova J, Jiao A, Lin L-L, Latz E, Hepworth D, Hall JP (2016) **Efficacy and Pharmacology of the NLRP3 Inflammasome Inhibitor CP-456,773 (CRID3) in Murine Models of Dermal and Pulmonary Inflammation** *J Immunol* **197**:2421–2433 <https://doi.org/10.4049/jimmunol.1600035>
- Resch W, Ziermann R, Parkin N, Gamarnik A, Swanstrom R (2002) **Nelfinavir-resistant, amprenavir-hypersusceptible strains of human immunodeficiency virus type 1 carrying an N88S mutation in protease have reduced infectivity, reduced replication capacity, and reduced fitness and process the Gag polyprotein precursor aberrantly** *J Virol* **76**:8659–8666 <https://doi.org/10.1128/jvi.76.17.8659-8666.2002>
- Richardson MW, Carroll RG, Stremlau M, Korokhov N, Humeau LM, Silvestri G, Sodroski J, Riley JL (2008) **Mode of Transmission Affects the Sensitivity of Human Immunodeficiency Virus**

Type 1 to Restriction by Rhesus TRIM5α *J Virol* **82**:11117–11128 <https://doi.org/10.1128/jvi.01046-08>

Rodriguez-Garcia M, Barr FD, Crist SG, Fahey JV, Wira CR (2014) **Phenotype and susceptibility to HIV infection of CD4+ Th17 cells in the human female reproductive tract** *Mucosal Immunol* **7**:1375–1385 <https://doi.org/10.1038/mi.2014.26>

Roesch F, OhAinle M, Emerman M (2018) **A CRISPR screen for factors regulating SAMHD1 degradation identifies IFITMs as potent inhibitors of lentiviral particle delivery** *Retrovirology* **15**:26 <https://doi.org/10.1186/s12977-018-0409-2>

Russell RA, Martin N, Mitar I, Jones E, Sattentau QJ (2013) **Multiple proviral integration events after virological synapse-mediated HIV-1 spread** *Virology* **443**:143–149 <https://doi.org/10.1016/j.virol.2013.05.005>

(no date) **Stanford - HIV Drug Resistance Database. n.d.** <https://hivdb.stanford.edu/page/release-notes/#consensusbsequences> <https://hivdb.stanford.edu/page/release-notes/#consensusbsequences>

Thompson MR, Sharma S, Atianand M, Jensen SB, Carpenter S, Knipe DM, Fitzgerald KA, Kurt-Jones EA (2014) **Interferon γ-inducible Protein (IFI) 16 Transcriptionally Regulates Type I Interferons and Other Interferon-stimulated Genes and Controls the Interferon Response to both DNA and RNA Viruses** *J Biol Chem* **289**:23568–23581 <https://doi.org/10.1074/jbc.M114.554147>

Toohey K, Wehrly K, Nishio J, Perryman S, Chesebro B (1995) **Human immunodeficiency virus envelope V1 and V2 regions influence replication efficiency in macrophages by affecting virus spread** *Virology* **213**:70–79 <https://doi.org/10.1006/viro.1995.1547>

Tsu BV, Agarwal R, Gokhale NS, Kulsuptrakul J, Ryan AP, Fay EJ, Castro LK, Beierschmitt C, Yap C, Turcotte EA, Delgado-Rodriguez SE, Vance RE, Hyde JL, Savan R, Mitchell PS, Daugherty MD (2023) **Host-specific sensing of coronaviruses and picornaviruses by the CARD8 inflammasome** *PLoS Biol* **21**:e3002144 <https://doi.org/10.1371/journal.pbio.3002144>

Varghese V, Mitsuya Y, Fessel WJ, Liu TF, Melikian GL, Katzenstein DA, Schiffer CA, Holmes SP, Shafer RW (2013) **Prototypical Recombinant Multi-Protease-Inhibitor-Resistant Infectious Molecular Clones of Human Immunodeficiency Virus Type 1** *Antimicrob Agents Chemother* **57**:4290–4299 <https://doi.org/10.1128/AAC.00614-13>

Vermeire J, Naessens E, Vanderstraeten H, Landi A, Iannucci V, Van Nuffel A, Taghon T, Pizzato M, Verhasselt B. (2012) **Quantification of reverse transcriptase activity by real-time PCR as a fast and accurate method for titration of HIV, lenti- and retroviral vectors** *PloS One* **7**:e50859 <https://doi.org/10.1371/journal.pone.0050859>

Walter BL, Wehrly K, Swanstrom R, Platt E, Kabat D, Chesebro B (2005) **Role of low CD4 levels in the influence of human immunodeficiency virus type 1 envelope V1 and V2 regions on entry and spread in macrophages** *J Virol* **79**:4828–4837 <https://doi.org/10.1128/JVI.79.8.4828-4837.2005>

Wang Q, Clark KM, Tiwari R, Raju N, Tharp GK, Rogers J, Harris RA, Raveendran M, Bosinger SE, Burdo TH, Silvestri G, Shan L (2024) **The CARD8 inflammasome dictates HIV/SIV**

pathogenesis and disease progression *Cell* **187**:1223–1237 <https://doi.org/10.1016/j.cell.2024.01.048>

Wang Q, Gao H, Clark KM, Mugisha CS, Davis K, Tang JP, Harlan GH, DeSelm CJ, Presti RM, Kutluay SB, Shan L (2021) **CARD8 is an inflammasome sensor for HIV-1 protease activity** *Science* **371**:eabe1707 <https://doi.org/10.1126/science.abe1707>

Wannamaker W, Davies R, Namchuk M, Pollard J, Ford P, Ku G, Decker C, Charifson P, Weber P, Germann UA, Kuida K, Randle JCR (2007) **(S)-1-((S)-2-[[1-(4-amino-3-chloro-phenyl)-methanoyl]-amino]-3,3-dimethyl-butanoyl)-pyrrolidine-2-carboxylic acid ((2R,3S)-2-ethoxy-5-oxo-tetrahydro-furan-3-yl)-amide (VX-765), an orally available selective interleukin (IL)-converting enzyme/caspase-1 inhibitor, exhibits potent anti-inflammatory activities by inhibiting the release of IL-1beta and IL-18** *J Pharmacol Exp Ther* **321**:509–516 <https://doi.org/10.1124/jpet.106.111344>

Xie M, Leroy H, Mascarau R, Woottum M, Dupont M, Ciccone C, Schmitt A, Raynaud-Messina B, Vérollet C, Bouchet J, Bracq L, Benichou S (2019) **Cell-to-Cell Spreading of HIV-1 in Myeloid Target Cells Escapes SAMHD1 Restriction** *mBio* **10** <https://doi.org/10.1128/mbio.02457-19>

Zhang C, Song J-W, Huang H-H, Fan X, Huang L, Deng J-N, Tu B, Wang K, Li J, Zhou M-J, Yang C-X, Zhao Q-W, Yang T, Wang L-F, Zhang J-Y, Xu R-N, Jiao Y-M, Shi M, Shao F, Sékaly R-P, Wang F-S (2021) **NLRP3 inflammasome induces CD4+ T cell loss in chronically HIV-1-infected patients** *J Clin Invest* **131** <https://doi.org/10.1172/JCI138861>

Zhong P, Agosto LM, Ilinskaya A, Dorjbal B, Truong R, Derse D, Uchil PD, Heidecker G, Mothes W (2013) **Cell-to-Cell Transmission Can Overcome Multiple Donor and Target Cell Barriers Imposed on Cell-Free HIV** *PLOS One* **8**:e53138 <https://doi.org/10.1371/journal.pone.0053138>

Kasai F, Hirayama N, Fukushima M, Kohara A, Nakamura Y (2022) **THP-1 reference data: Proposal of an in vitro branched evolution model for cancer cell lines** *Int J Cancer* **151**:463–472 <https://doi.org/10.1002/ijc.34019>

Kulsuptrakul J, Turcotte EA, Emerman M, Mitchell PS (2023) **A human-specific motif facilitates CARD8 inflammasome activation after HIV-1 infection** *eLife* **12**:e84108 <https://doi.org/10.7554/eLife.84108>

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Joint Public Review:

Following up on their previous work, the authors investigated whether HIV-1 cell-to-cell transmission activates the CARD8 inflammasome in macrophages, a key question given that inflammasome activation in myeloid cells triggers proinflammatory cytokine release. Co-cultures of HIV-infected T cells with macrophages led to viral spreading, resulting in IL1 β release and cell death, with CARD8 playing a crucial role in this inflammasome response, triggered by HIV protease. The authors also found that HIV isolates resistant to protease inhibitors showed differences in CARD8 activation and IL1 β production, highlighting the clinical relevance of their findings. Overall, this well-written study provides strong evidence for the role of CARD8 in protease-dependent sensing of viral spread, with implications for understanding chronic inflammation in HIV infections and its potential contribution to systemic immune activation, especially under ART. The authors have addressed initial weaknesses and verified effects in cocultures of primary T cells and macrophages. They now also provide evidence that CARD8 is activated by protease from incoming viral particles. Further studies are needed to clarify how much this mechanism contributes to systemic immune activation in untreated infections and whether this mechanism drives chronic inflammation under ART.

<https://doi.org/10.7554/eLife.102676.2.sa1>

Author response:

The following is the authors' response to the current reviews.

We again thank you for the positive and constructive feedback on our manuscript, and for highlighting its contributions to understanding the role of CARD8 in viral protease-triggered sensing of viral spread, and the potential impact of our findings on chronic inflammation and immune activation. We agree that it will be important for future work to address whether or not HIV-1 protease-triggered CARD8 inflammasome activation contributes to chronic inflammation in PLWH who are receiving ART.

In response to the question about the baseline level of IL-1 β in Fig. 4D, the figure below shows the mock condition for the CD4 $^{+}$ T cell:MDM coculture. We had done this control in parallel with the data presented in the submitted figure. Levels of IL-1 β during HIV-1 infection are increased over background (i.e., mock infection). We note that for donor G the

IL-1 β concentration is below the limit of detection for this assay. Thus, it remains possible that other inflammasomes contribute modestly during cell-to-cell transmission of HIV-1; however, incomplete knockout of *CARD8* in a minority of cells may also contribute to the observed levels of IL-1 β in response to HIV-1 infection. Nonetheless, collectively, our data strongly supports the role for *CARD8* in HIV-1 protease-triggered inflammasome activation.

The following is the authors' response to the original reviews.

Joint Public Review:

Following up on their previous work, the authors investigated whether cell-to-cell transmission of HIV-1 activates the CARD8 inflammasome in macrophages, an important question given that inflammasome activation in myeloid cells triggers proinflammatory cytokine release. The data support the idea that CARD8 is activated by the viral protease and promotes inflammation. However, time-course analyses in primary T cells and macrophages and further information on the specific inflammasome involved would further increase the significance of the study.

Strengths:

The manuscript is well-written and the data is of good quality. The evidence that CARD8 senses the HIV-1 protease in the context of cell-to-cell transmission is important since cell-to-cell transmission is thought to play a key role in viral spread in vivo, and inflammation is a major driver of disease progression. Clean knockout experiments in primary macrophages are a notable strength and the results clearly support the role of CARD8 in protease-dependent sensing of viral spread and the induction of IL1 β release and cell death. The finding that HIV-1 strains are resistant to protease inhibitors differ in CARD8 activation and IL1 β production is interesting and underscores the potential clinical relevance of these results.

Weaknesses:

One weakness is that the authors used T cell lines which might not faithfully reflect the efficiency of HIV-1 production and cell-cell transfer by primary T cells. To assess whether CARD8 is also activated by protease from incoming viral particles earlier time points should be analyzed. Finally, while the authors exclude the role of NLRP3 in IL-1b and the death of macrophages it would be interesting to know whether the effect is still Gasdermin D dependent.

Recommendations for the authors

(1) Co-culture assay should also be done between primary CD4 cells and primary MDMs, because T-cell lines produce much more viruses, and the efficiency of cell-to-cell transmission might be dramatically different in primary cells compared to cell lines.

We have now added data from experiments using infected primary CD4 cells as the donor cells in cell-to-cell HIV-1 transmission to MDMs in new Figure 4. The results largely phenocopy the SUPT1:MDM coculture in that we observe inflammasome activation after co-culture of HIV-infected primary T cells with primary MDMs. We find that this inflammasome activity induced by the CD4:MDM cell-to-cell transmission is abrogated by knockout of *CARD8* in the MDMs or treatment of HIV protease inhibitor lopinavir (LPV) or caspase 1 inhibitor VX765, suggesting that this activation is dependent on *CARD8*, HIV protease, and caspase 1. Additionally, the signal persists in the presence of reverse transcriptase inhibitor nevirapine (NVP), suggesting that the incoming protease is driving activation.

(2) For all co-culture experiments, supernatants were collected at 48 or 72 hours. Since CARD8 activation is expected to be driven by incoming viral particles without RT, they should measure cytokine production at much earlier time points. 2-3 days co-culture raises concerns. Ideally, the authors can provide a time-course.

We have now added a time course of the SUPT1:MDM coculture from 3 unique donors taken at 4, 24, 48, and 72 hours post coculture in the presence or absence of reverse transcriptase inhibitor (see new Figure 3B) as well as for the primary CD4 cells to MDM co-culture (see new Figure 4B). We detect IL-1 β at the 24hour time point (and later), but not at the 4-hour time point which is slower than what was detected by direct cell-free infection (Kulsuptrakul et al., 2023). However, we still hypothesize that this is driven by active incoming viral protease because the signal is not abrogated by a reverse transcriptase inhibitor, which indicates that *de novo* protease production is not necessary. We also observed that IL-1 β levels do not increase after plateauing 24h after establishing the co-culture, suggesting that secondary infection does not further amplify inflammasome activation. We now speculate on this in the Discussion.

(3) A potential confounder in the data in Figure 4 is that despite rightly including the cognate adaptations in the Gag cleavage sites with the PI-R protease mutants, some of these viruses still display Gag processing defects. Can the authors disentangle the potency of PR mutant cleavage with either reduced cell entry or reduced protease availability due to processing defects in the incoming virions?

The reviewer is correct that although the western blot with the p24^{gag} antibody suggests that Gag is processed, we cannot rule out that other variables do not contribute to the observed difference in CARD8 inflammasome activation. For example, PI-R clones relative to the LAI strain may have distinct protease substrate specificity, variable efficiency/kinetics in viral assembly, gag dimerization, and other factors may ultimately influence CARD8 inflammasome activation. We have updated the text to reflect these possibilities. Nonetheless, this argument does not change the conclusion that CARD8 inflammasome activation is affected by protease mutations acquired during drug resistance.

(4) There is considerable donor variation in the macrophages (unsurprising) but can the authors correlate this with CARD8 expression and are there any off-target effects on macrophage permissivity to HIV-1 infection?

We have now considerably increased the number of primary cell donors from the first submission (see Author response table 1 below). We find that the non-responsive donor presented in the first submission is aberrant since all others do respond to a greater or lesser degree (Figure 3, Figure 4). However, the reviewer may be correct that the particular aberrant donor MDMs were poorly infected. We also note that despite donor variability in the degree of activation (IL-1 β secretion) from cocultures with HIV_{BaL}-infected SUPT1 cells, HIV-induced activation is comparable to the activation induced by VbP (see new Figure 3-figure supplement 1B). We do not see a notable difference in CARD8 expression between donors. Nonetheless, with the added number of primary cell donors, the data are consistent with a role of primary MDMs from nearly all donors in supporting a CARD8-dependent, HIV-protease dependent inflammasome response after co-culture with infected T cells. We have left in data from all of the donors so that readers can appreciate the variability among primary cells.

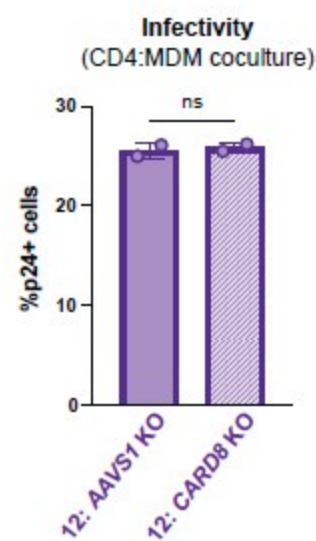
Author response table 1.

	Preprint	Resubmission
# of WT MDM donors	3 (Figure 3A)	9 (Figure 3A and B, Figure 4A and B)
# of KO MDM donors	3 (Figure 3C and D: donor 7-9)	7 (Figure 3C and D, Figure 4C and D donor 7-12,15)
# of SUPT1:MDM donors	6 (Figure 3)	11 (Figure 3)
# of CD4:MDM donors	0	4 (Figure 4)

In addition, to address the reviewer concerns about off-target effects of the sgRNAs on macrophage permissivity, we assessed our CD4:MDM cocultures for percent infectivity via intracellular p24^{gag} in AAVS1 vs CARD8 KO MDMs and we observed no significant difference in infectivity in AAVS1 vs CARD8 KO MDMs (see Author response image 1 of MDMs after co-culture with T cells that is not affected any potential off-target effects of the sgRNAs.

Author response image 1.

Equivalent infection in AAVS1 vs CARD8 KOMDMs. AAVS1 or CARD8 KO from donor 12 were cocultured with mock or HIV infected CD4 T cells as described in Figure 4D for 72 hours then assessed for HIV infection of the MDMs by washing away CD4 T cells, harvesting MDMs, and staining attached MDMs for intracellular p24^{gag} for flow cytometry analysis. Datasets represent mean ± SD (n=2 technical replicates from one donor). One-way ANOVA with Dunnett’s test using GraphPad Prism 10. ns = not significant, *p<0.05,**p<0.01, ***p<0.001, ****p<0.0001.



(5) The authors suggest that NLRP3 is unlikely to be the mediator of IL-1b and cell death in the macrophages. Is this death still GSDMDdependent, what other NLRs are expressed in this system and does it make a difference what PAMP you use to prime the response?

We have now added additional data in support of the conclusion that NLRP3 is not a mediator of the IL-1β secretion in the infected SUPT1 cells to primary MDMs coculture. In addition to using an NLRP3 inhibitor, we have now also made NLRP3 KOs MDMs and used these in the

coculture experiments which show that the IL-1 β secretion after coculture of infected SUPT1 cells and primary MDMs is mediated by CARD8 and not NLRP3 because the signal is abrogated by CARD8 knockout, but not by NLRP3 knockout. This new data is shown in Figure 3C and D.

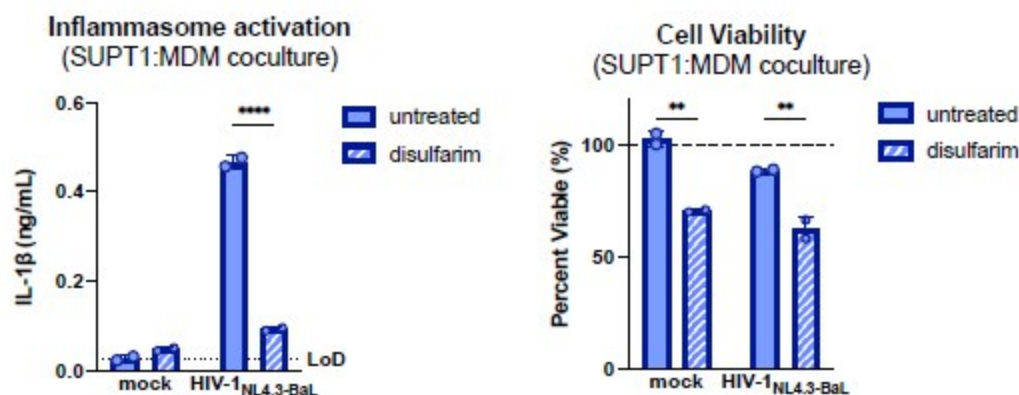
To assess the role of GSDMD, we treated SUPT1:MDM cocultures with disulfiram, a GSDMD inhibitor (Hu et al., 2020). Disulfiram treatment abrogated IL-1 β secretion, suggesting that this activation is indeed GSDMD-mediated (see Author response image 2 below). We choose not to include the disulfiram result in the final manuscript since we have not ruled out cytotoxic effects of the drug.

There are likely other NLRs expressed in primary MDMs; however, since inflammasome activation is completely absent in the *CARD8* KO MDMs, we infer that *CARD8* is the main inflammasome-forming sensor in this system. However, we cannot rule out the possibility of other innate sensors being activated downstream of *CARD8* or under different differentiation conditions.

To address the concern that alternative priming affects *CARD8* activation, we compared pre-treatment of cells with Pam3CSK4 or lipopolysaccharide (LPS) in the presence or absence of HIV protease inhibitor and reverse transcriptase inhibitor. Regardless of the priming agent used, we observed HIV protease-dependent activation that persisted in the presence of reverse transcriptase inhibitor, suggesting that *CARD8* is the main sensor under LPS and Pam3CSK4 priming (new Figure 3–figure supplement 1A).

Author response image 2.

Inflammasome activation following cell-to-cell HIV infection is mediated by GSDMD. SUPT1-CCR5 cells were either mock-infected or infected with HIV-1_{NL4.3BaL} for 20 hours before coculturing with MDMs in either the presence or absence of GSDMD inhibitor disulfiram (25 μ M). Cocultures were harvested 24 hours later to assess (left) IL-1 β secretion via IL-1 reporter assay and (right) cell viability via CellTiter-Glo® assay. Viability was calculated by normalizing to relative luminescence units in the mock untreated control. Dotted line indicates limit of detection (LoD). Dashed line indicates 100% viability as determined by untreated mock control. Datasets represent mean $\pm$ SD (n=2 technical replicates for one donor). Two-way ANOVA with Sidak's test (using GraphPad Prism 10. ns = not significant, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001).



Minor points

(1) In Figure 1, the authors should clarify whether LAI or LAI-VSV-G was used.

Wild-type virus (LAI strain) was used in Figure 1. This has now been clarified in the figure legend.

(2) In Figure 1, the fraction of infected cells without DEAE was ~20% in both WT and CARD8 KO THP-1, suggesting somewhat efficient viral entry even in the absence of DEAE. How do the authors reconcile this with the lack of IL-1 β production? The increase in infection observed in WT THP-1 +DEAE was overall modest (from ~20% to 25-30%) compared to the dramatic difference in IL-1 β production. Can they provide more evidence or discuss how DEAE might be impacting cytokine production? If differences in viral entry are the explanation for differences in inflammasome activation, then they should be able to overcome this by using virus at a higher MOI in the absence of DEAE. Experiments proposed in Figure 1 +/- DEAE should be repeated using a range of MOI for LAI and showing the corresponding percent infection in THP-1 cells (which is not shown in Figure S2 for LAI-VSVG).

We hypothesize that the lack of IL-1 β production without DEAE is likely due to an insufficient amount of incoming viral protease to induce CARD8 activation. Though the increase in infection with DEAE is modest by intracellular p24^{gag} at 24 hours post infection, we infer that intracellular p24^{gag} may be largely underestimating the actual increase in viral efficiency achieved with DEAE (now in Supplemental Note). We have also updated Figure S2 (now Figure 2—figure supplement 1) legend to include the percent infection for HIV-1_{LAI} and HIV-1_{LAI-VSVG} infections. We agree that activation in the absence of DEAE could be overcome by infecting with a more concentrated viral stock to increase the MOI. Indeed, our decision to use the cell-to-cell transmission model achieves this in a more physiologic context.

(3) In Figure S1, the authors point out that RT-activity in the supernatants was similar in the cell-free vs. cell-to-cell model. While in the transwell system THP-1 cells are the only cells capable of producing new virions, how are they able to differentiate viral production from sup-T1 vs. THP-1 in the cell-to-cell system? At a minimum, they should provide some data on the observed RT activity in matching wells containing the same number of infected sup-T1 cells utilized in coculture experiments.

We think this may have been a misinterpretation. In Figure S1 (now Figure 1B, right), we compare the amount of virus available in the lower chamber of the transwell versus the cell-to-cell condition. We are not comparing cell-free to cell-to-cell infection. We have changed the text and figure title to clarify this point.

(4) Can the authors provide additional comments on the lack of IL-1 β release in donor C in Figure 3? The donor did not produce IL-1 β in response to VbP or HIV, although the WB for CARD8 appears similar to the other two donors.

We have now tested MDMs from additional donors and continue to find a range of IL-1 β secretion after the coculture. However, donor C is aberrant since each of the other donors had detectable IL-1 β secretion in response to VbP and HIV-1 to greater or lesser extents. Nonetheless, we have included additional donors summarized in the table above corresponding to major comment #4.

(5) For Figure 3, can the authors provide information on the fraction of MDMs that were infected after coculture with sup-T1 cells? Why didn't the authors measure cell death in MDMs?

It is difficult to measure the fraction of MDMs infected or dying in the cocultures since it is hard to separate signal from the T cells. Although it would be possible to do so, in this manuscript, we instead prefer to focus on the potential contribution of CARD8 inflammasome

activation in exacerbating chronic inflammation in response to HIV rather than the depletion of macrophages.

(6) In Figure 4, did the authors introduce the mutations associated with PI resistance into the same LAI backbone? If not, this is not a fair comparison, as viral protein expression levels were not at the same level, indicated in Figure 4A. Additionally, such comparison will be further strengthened by using cells other than 293T cells for the coculture assay.

No, we did not introduce these mutations into LAI, since they were already in an NL4.3 backbone and NL4.3 and LAI differ by only 1 amino acid in protease. We have updated Table S1 to report this amino acid difference. We also note that in our previous manuscript we tested much more diverse proteases such as a clade A HIV-1, HIV-2, and SIVs and find comparable CARD8 cleavage to LAI.

Additions not requested by Reviewers:

THP-1 characterization

In our previous work, we noticed that different “wildtype” THP-1 lines behaved uniquely in response to DEAE-dextran. In particular, we observed inflammasome activation in response to DEAE-dextran alone at the concentration used for spinoculations (20µg/mL), whereas the other THP-1 line did not. Thus, we performed STR profiling on each THP-1 cell line and determined that the THP-1 cells used in our studies (JK THP1s) are distinct from THP-1 cells from ATCC at 3 different loci. This data is now included in the Supplemental Note (Figure A1). Please note that all data in this and the accompanying manuscript were performed in JK THP-1 cells.

Whole plasmid sequencing of the PI-resistant HIV clones

Since preprint submission, we have done whole plasmid Oxford Nanopore sequencing on the PI-resistant HIV clones obtained from the NIAID HIV/AIDS Specimen Repository Program. Of note, there were a handful of previously unreported mutations included in these plasmid stocks within protease. We have updated Table S1 to include an additional column titled “Additional amino acid changes in HIV^{PR} relative to NL4.3.”

References

Hu JJ, Liu X, Xia S, Zhang Z, Zhang Y, Zhao J, Ruan J, Luo X, Lou X, Bai Y, Wang J, Hollingsworth LR, Magupalli VG, Zhao L, Luo HR, Kim J, Lieberman J, Wu H. 2020. FDA-approved disulfiram inhibits pyroptosis by blocking gasdermin D pore formation. *Nat Immunol* 21:736–745. doi:10.1038/s41590-020-0669-6

Kulsuptrakul J, Turcotte EA, Emerman M, Mitchell PS. 2023. A human-specific motif facilitates CARD8 inflammasome activation after HIV-1 infection. *eLife* 12:e84108. doi:10.7554/eLife.84108

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